Kinetic Theory of Gases

Pressure of an ideal gas:

* Ideal gas — which obeys gas laws at all temps.

Assumptions :-

① All molecules of the gas are pt sized (hard spheres)

② They move at random in all directions at any temp. So they collide with each other and also with walls of container.

③ all collisions are perfectly elastic.

④ The collision b/w the molecules are neglected. The time of collision is negligible when ~~compared~~ compared to the time b/w the collision.

⑤ Between two successive collisions, molecules move along st. line path.

⑥ Molecular collisions obey "newtons laws".

⑦ The volume of gas is equal to the volume of container enclosed by gas. but vol of gas is not equal to volume of all molecules of gas.

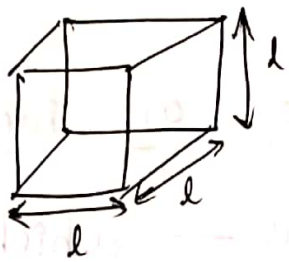
⑧ Mass of gas is equal to mass of all molecules of gas.

Derivation :-

let us consider a cubical vessel,

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Consider a gas enclosed in a closed cubical vessel of side length l.



Volume of gas

V = Vol of vessel

$V = l^3$ — (1)

m_0 = mass of each gas molecule.

N = Total no. of molecules of gas.

m = mass of gas = $m_0 \times N$.

$\rho = \frac{m}{V}$ = density of the gas.

Let $\bar{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$ is the rms speed of a molecule.

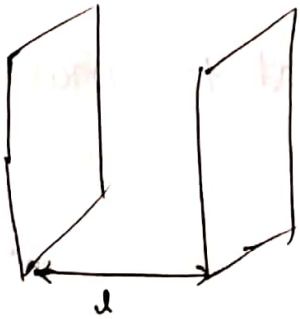
$$v^2 = v_x^2 + v_y^2 + v_z^2 \quad -(2)$$

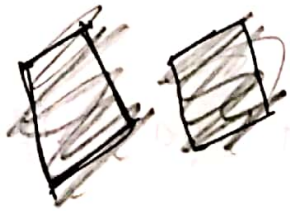
by probability,

$$v_x^2 = v_y^2 = v_z^2$$

eqn (2) becomes $\boxed{v_x^2 = v_y^2 = v_z^2 = \frac{v^2}{3}}$.

*





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* momentum of a molecule just before collision $= m_0 v_x$

* and just after collision $= -m_0 v_x$

* change in momentum of molecules in one molecule in each collision $= -m_0 v_x - m_0 v_x = -2m_0 v_x$

* time interval b/w two successive collisions of a molecule on the same wall,

$$\Delta t = \frac{2l}{v_x}$$

* collision frequency $= \frac{1}{\Delta t}$ = no. of collisions per second.

* Avg force on the wall $= \frac{(\Delta P)_{wall}}{\Delta t} = \frac{-(-2m_0 v_x)}{\frac{2l}{v_x}}$

$$\boxed{\text{Avg force} = \frac{m_0 v_x^2}{l}}$$

* Avg force on wall due to N-molecules of gas,

$$F_x = \frac{N m_0 v_x^2}{l} = \frac{m_0 v^2}{3l}$$

* pressure of gas, $p = \frac{F_x}{A} = \frac{mv^2}{3l(l^2)} = \frac{mv^2}{3l^3}$

$$p = \frac{mv^2}{3V}$$

$$\boxed{PV = \frac{1}{3} m v_{rms}^2}$$

$$P = \frac{1}{3} \times \frac{m}{V} \times v_{rms}^2 \Rightarrow \boxed{P = \frac{1}{3} \rho v_{rms}^2}$$

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Imp point:

(1) n_0 = molecular density = $\frac{N}{V}$

N = no. of molecules of gas

V = volume of gas.

by probability,

No. of molecules travelling towards each wall

$$= \frac{N}{6} = \frac{m_0 V}{6}$$

(2) Translational K.E of each molecule is,

$$K.E = \frac{1}{2} m_0 v_{rms}^2 \quad - (1)$$

for a gas,

$$K.E = \frac{1}{2} m v_{rms}^2 \quad - (2)$$

we have,

$$PV = \frac{1}{3} m v_{rms}^2$$

$$PV = \frac{2}{3} \times \frac{1}{2} m v_{rms}^2$$

$$\boxed{PV = \frac{2}{3} K.E}$$

$$P = \frac{2}{3}\left(\frac{K.E}{V}\right)$$

$$\boxed{P = \frac{2}{3}(E)}$$

$$\epsilon = \frac{E.E}{V} = K.E \text{ per unit vol of gas.}$$

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) Ideal gas eqⁿ,

$$PV = nRT$$

no. of moles, $n = \frac{m}{M} = \frac{N}{N_A}$

$$\boxed{N = \frac{mN_A}{M}}$$

$N_A = 6.023 \times 10^{23}$ atoms/gm mole

m = mass of gas in gms

M = molecular wt of gas in gms.

and $m = m_0 N$

$$\frac{m_0 N N_A}{M} = N$$

$$\boxed{m_0 = \frac{M}{N_A}}$$

m_0 — mass of molecule

*Again,

$$\boxed{R = N_A \times K_B}$$

K_B — Boltzmann's const $= 1.38 \times 10^{-23}$ J/K

4)

$$PV = nRT$$

$$\frac{1}{3} m v_{rms}^2 = nRT$$

$$v_{rms}^2 = \frac{3}{M} \frac{RT}{n}$$

$$\boxed{v_{rms} = \sqrt{\frac{3RT}{M}}}$$

M – molecular wt.

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* $V_{rms} = \sqrt{\frac{3RT}{M}}$

$= \sqrt{\frac{3 \times N_A K_b \times T}{m_0 N_A}}$

$$\boxed{V_{rms} = \sqrt{\frac{3K_bT}{m_0}}}$$

(5) Translational K.E of a molecules,

$$K.E = \frac{1}{2} m_0 v_{rms}^2$$

$$= \frac{1}{2} \times m_0 \times \frac{3K_bT}{m_0}$$

$$\boxed{K.E = \frac{3}{2} K_bT}$$

(6) Translational K.E of a gas,

$$K.E = \frac{3}{2} N K_b T$$

(7) Translational K.E of 1 mole of gas,

$$K.E = \frac{3}{2} N_A K_b T$$

$$\boxed{K.E = \frac{3}{2} RT}$$

(8) Translational K.E of n moles of gas,

$$\boxed{K.E = \frac{3}{2} nRT}$$

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Notes-

① At a given temp, Translational K.E is same for all molecules of different gases. becoz T.K.E depends on T but not on m.

② rms speed is

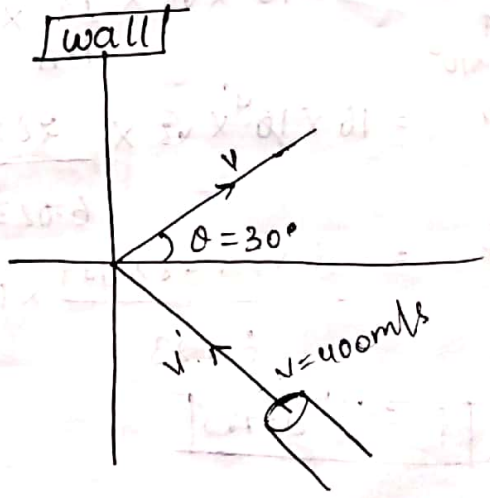
$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$V_{rms} \propto \sqrt{T}$ (when M = const)

$V_{rms} \propto \frac{1}{\sqrt{M}}$ (when T = const)

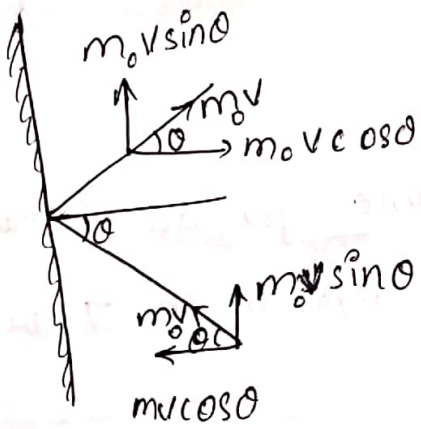
* $M_{O_2} > M_{N_2} \Rightarrow (V_{rms})_{O_2} < (V_{rms})_{N_2}$ (at T = same)

Ex:
A parallel beam of N_0 molecules moving with velocity $v = 400 m/s$ impinges (strikes) on a wall at angle $\theta = 30°$ to its normal. The conc of molecules in the beam $n_0 = 0.9 \times 10^{19}/cm^3$. Find the pressure exerted by the beam on the wall. Assuming collisions of molecules with the wall are perfectly elastic.



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Sol:



change in momentum of each molecule

$= 2 m_0 v \cos\theta$

$= 2 \times m_0 \times 400 \times \frac{\sqrt{3}}{2}$

$= 400\sqrt{3}\, m_0$

force on the wall = $\Delta P \times$ collision frequency

$= 400\sqrt{3} m_0 \times \frac{n_0 v}{t}$

$F = 2 m_0 v \cos\theta \times \frac{n_0 v}{t}$

$= 2 m_0 n_0 v^2 \cos\theta \cdot A$

$\frac{F}{A} = 2 m_0 n_0 v^2 \cos\theta$

$P = 2 m_0 n_0 v^2 \cos\theta$

$= 2 \times \frac{M}{N_A} \times n_0 v^2 \cos\theta$

$= 2 \times \frac{28 \times 10^{-3} \times 0.9 \times 10^{19} \times 10^6}{6.023 \times 10^{23}} \times 400 \times \frac{3}{4}$

$= 1159 \times 10^{-3}$

$P = 1.15 \times 10^5$ Pa

$\approx$ P = 1 atm

$= 400\sqrt{3} m_0 \times \frac{n_0 v}{t}$

$= 400\sqrt{3} m_0 \times n_0 \times \frac{A\ell}{t} \times v$

$= 400\sqrt{3} m_0 \times n_0 \times A \times 400$

$F = 160000\sqrt{3} m_0 \times 10^2 \times A \times n_0$

$\frac{F}{A} = 160000\sqrt{3} m_0 \times n_0$

$P = 16 \times 10^4 \times \sqrt{3} \times \frac{M}{N_A} \times n_0$

$= 16 \times 10^4 \times \sqrt{3} \times \frac{28 \times 10^{-3}}{6.023 \times 10^{23}} \times 0.9 \times 10^{19}$

$= \frac{16\sqrt{3} \times 28 \times 0.9}{6.023} \times 10^{-3} \times 10^6$

P = 1 atm

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Ex: One gmmole of O_2 at 27°C & 1atm is enclosed in a vessel. Assuming the molecules to be moving with v_{rms}. Find the no. of collisions per second which the molecules make with one sqm area of the vessel wall.

$$N' = \frac{N}{6At} \quad - (1)$$

we have, $N = \frac{n_0 V}{6}$

$$N' = \frac{n_0 V}{6At}$$

$$= \frac{n_0 Ad}{6At}$$

$$\boxed{N' = \frac{n_0 v_{rms}}{6}}$$

Again,

$$PV = nRT$$

$$PV = \frac{N}{N_A} \cdot N_A \times k_b T$$

$$P = \frac{N}{V} k_b T$$

$$\boxed{n_0 = \frac{P}{k_b T}}$$

$$N' = \frac{P v_{rms}}{6 k_b T} = \frac{10^5 \sqrt{\frac{3RT}{M_0}}}{6 k_b T} = \frac{10^5 \sqrt{\frac{3 \times 8.314 \times 300}{32}}}{6 \times 1.38 \times 10^{-23} \times 300}$$

$$N' = 1.97 \times 10^{27} \, m^{-2} s^{-1}$$

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Eg:1) In a container, the molecular density of an enclosed gas is 10^{26} molecule/m^3 each of mass 3×10^{-27} kg. If V_{rms} of gas molecules is 2000 m/sec Find no. of molecules hitting per sq.m of containg walls every sec & The pressure exerted on the walls of container by the molecules is

2) The avg translational K.E of O_2 at a particular temp is 0.04 eV. Find avg trans. K.E of the molecules in e.V at same temp is 0.04 eV

1) $$N' = \frac{N}{6At} = \frac{n_0 V_{rms}}{6}$$

$$N' = \frac{10^{26}\times 2000}{6} = \frac{20}{6}\times 10^{28} = \frac{10}{3}\times 10^{28}$$

$$N' = 3.33\times10^{28}\ m^{-2}s^{-1}$$

$$PV = \frac{1}{3} m V_{rms}^2$$

$$P = \frac{1}{3}\frac{N m_0}{V} V_{rms}^2$$

$$P = \frac{1}{3}\left(\frac{N}{V}\right) m_0 V_{rms}^2$$

$$P = \frac{1}{3} n_0 m_0 V_{rms}^2$$

$$P = \frac{1}{3}\times10^{26}\times3\times10^{-27}\times4\times10^6 = 4\times10^5\ atm$$

$$P = 4\ Pa$$

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2) due to temp is const there will be no change in avg trans. K.E because it depends on on temp.

∴ avg tras. k. E = 0. ous ev

Gas Laws:

① Boyle's Law:

we have,

$$PV = \frac{1}{3} m v_{rms}^2$$

where, $v_{rms}^2 = \frac{3RT}{M}$

$$PV = \frac{1}{3} m \times \frac{3RT}{M}$$

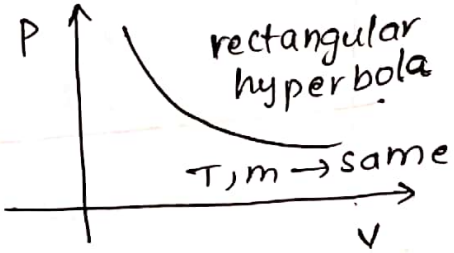
$$\boxed{PV = \frac{m}{M} RT}$$

for a given mass of a gas and at const temp T.

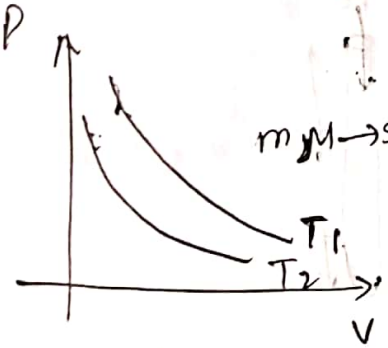
$$PV = \text{const}$$

i.e, $\boxed{V \propto \frac{1}{P}}$

* P

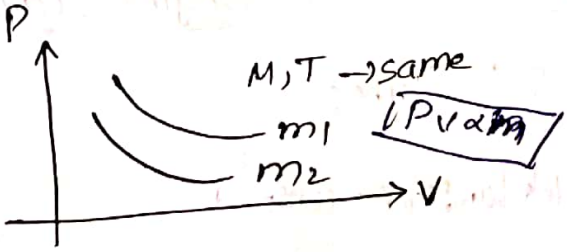


* P



clearly $T_1 > T_2$

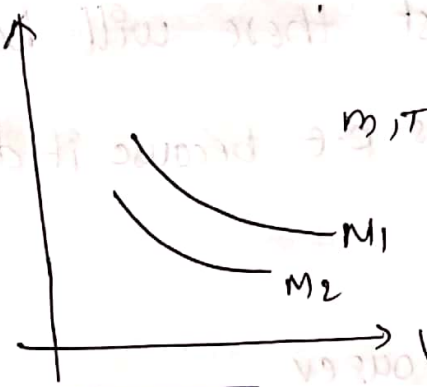
* P



clearly $m_1 > m_2$

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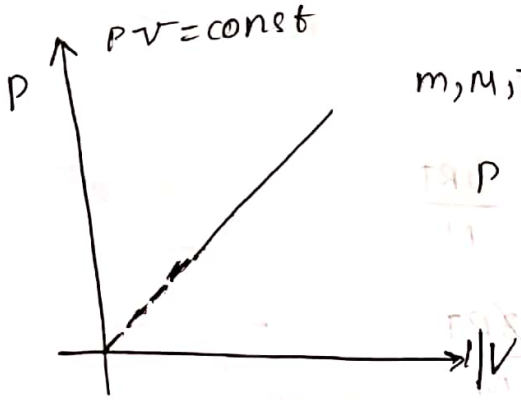
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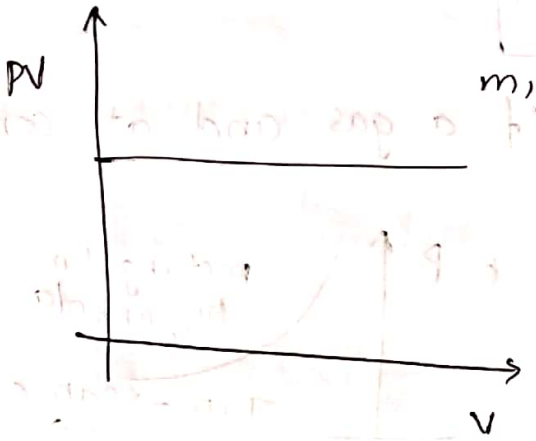
$PV \propto \frac{1}{M}$

$\frac{1}{M_1} > \frac{1}{M_2} \Rightarrow M_1 < M_2$

*

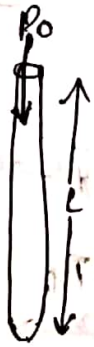


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Applications of Boyle's Law

1) Quill Tube:



boyle's law $P_0 V = \text{const}$

$P_0 A l = \text{const}$

$P_0 l = \text{const}$

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$\Rightarrow P_0 = Hsg = H\,cm$ of mercury.

$P_0 = 76\,cm$ of $H\cdot g$

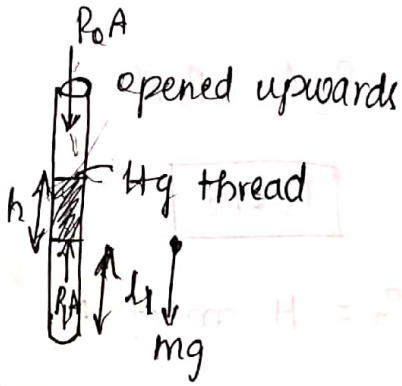
$= \frac{76}{100} \times 13.6 \times 10^3 \times 9.8\ N/m^2$

$= 76 \times 136 \times 9.8$

$\boxed{P = 1.013 \times 10^5\ N/m^2}$

$\Rightarrow \boxed{HL = const}$

now, ① since thread is in eqm,



$P_0A + mg = P_1A$

$P_0A + Ahsg = P_1A$

$\boxed{P_1 = P_0 + hsg}$

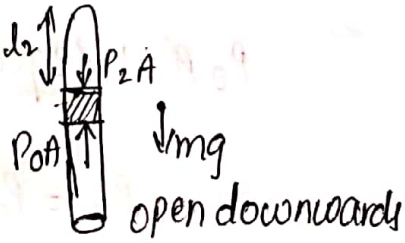
$P_1 = H + hsg$

$\boxed{P_1 = (H+h)sg}$

$P_1 = (H+h)$ cm of Mercury.

Boyle's law, $\boxed{(H+h)L = const}$

②



we have,

$P_2A + mg = P_0A$

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$P_2 A + h s g A = P_0 A$

$P_2 = P_0 - h s g$

$P_2 = H s g - h s g$

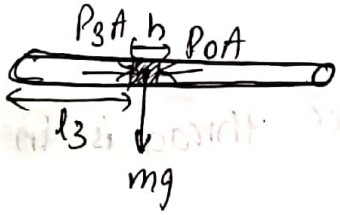
$$\boxed{P_2 = (H-h) s g}$$

$P_2 = (H-h)$ cm of Hg

$P_2 l_2 = \text{const}$

$$\boxed{(H-h) l_2 = \text{const}}$$

③



$P_3 A = P_0 A$

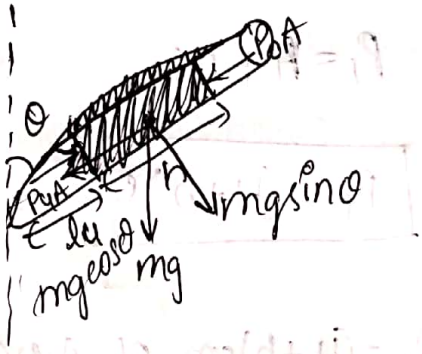
$$\boxed{P_3 = P_0}$$

$P_3 = H$ cm of Hg

$\Rightarrow P_3 l_3 = \text{const}$

$$\boxed{H l_3 = \text{const}}$$

④



we have boyle's law only for trapped air.

In eqm,

$P_0 A + mg\cos\theta = P_4 A$

$P_0 A + hAsg\cos\theta = P_4 A$

$P_4 = P_0 + h s g \cos\theta$

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$$P_4 = (H + h\cos\theta)\, s_g$$

$P_4 = (H + h\cos\theta)$ cm of Hg.

we have,

$$P_4 d_4 = \text{const}$$

$$\boxed{(H + h\cos\theta)\, d_4 = \text{const}}$$

Finally, $\boxed{(H+h)\, d_1 = (H-h)\, d_2 = H d_3 = (H + h\cos\theta)\, d_4 = \text{const}}$

Points:

① $(H+h) d_1 = (H-h) d_2$

$$H d_1 + h d_1 = H d_2 - h d_2$$

$$h(d_1 + d_2) = H(d_2 - d_1)$$

$$\boxed{h = \frac{H(d_2 - d_1)}{d_1 + d_2}}$$

② we have,

$$H d_3 = (H+h)\, d_1$$

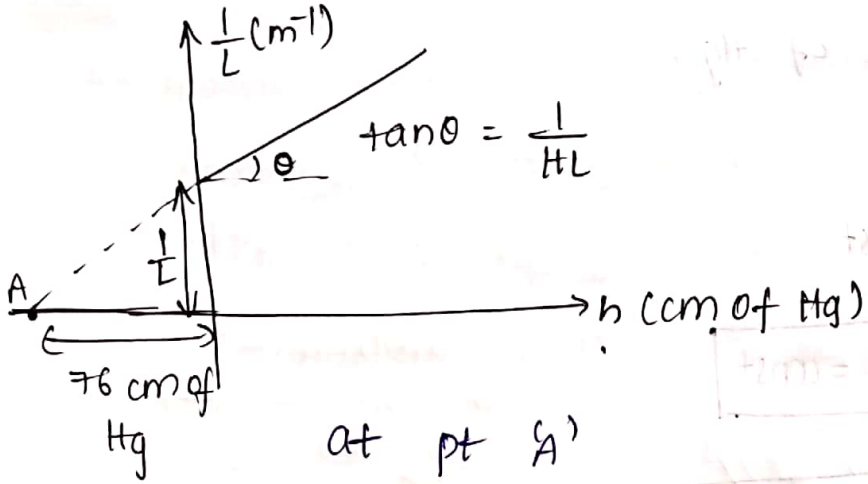
$$\frac{1}{d_1} = \frac{H+h}{H d_3}$$

$$\frac{1}{d_1} = \frac{H}{H d_3} + \frac{h}{H d_3}$$

$$\frac{1}{d_1} = \frac{h}{H d_3} + \frac{1}{d_3}$$

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$y = mx + c \Rightarrow \frac{1}{d} = \frac{h}{HL} + \frac{1}{L}$



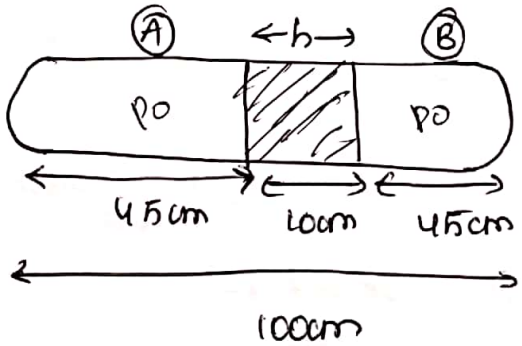
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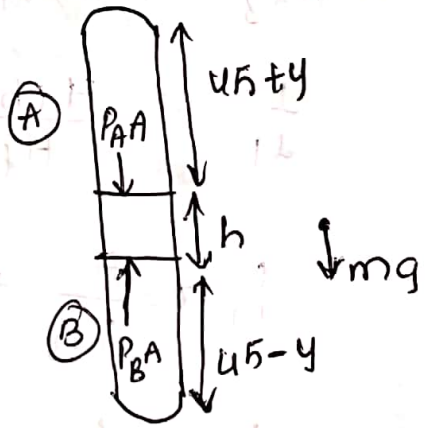
$\frac{1}{d} = 0 \Rightarrow \frac{h}{HL} + \frac{1}{L} = 0$

$\frac{h}{HL} = -\frac{1}{L}$

$\boxed{h = -H}$

Ex:
A column of Hg of 10cm length is contained in the middle of a narrow horizontal 1m long tube which is closed at both the ends both the halves of the tube contains air at a pressure of 76cm of Hg by what distance will column of Hg be placed if the tube is held vertical.





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Let y be the displacement of thread.

Thread is in eqm,

$$F_{net} = 0$$

$$P_A A + mg = P_B A$$

$$P_A A + h\rho Ag = P_B A$$

$$\Rightarrow P_B - P_A = h\rho g$$

$$P_B - P_A = 10 \quad -①$$

from Boyle's law,

$$PL = \text{const}$$

$$P_0(45) = P_A(45+y) \quad -① \Rightarrow P_A = \frac{P_0(45)}{45+y}$$

$$P_0(45) = P_A(45-y) \quad -② \Rightarrow P_B = \frac{P_0 \times 45}{45-y}$$

$$\frac{P_A}{P_B} = \frac{45-y}{45+y}$$

$$\frac{45P_0}{45-y} - \frac{P_0 \times 45}{45+y} = 10$$

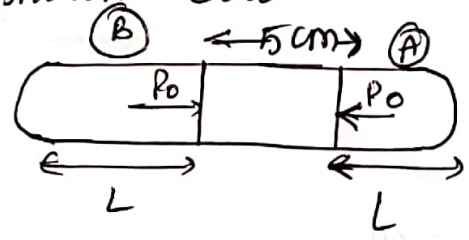
$$45P_0\left(\frac{45+y-45+y}{(45-y)(45+y)}\right) = 10$$

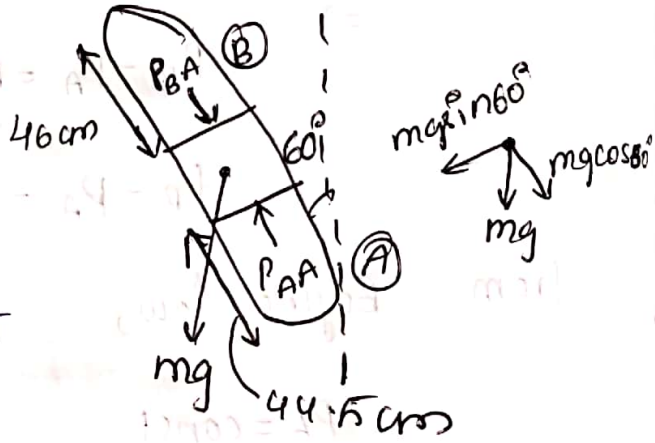
$$90P_0 y = 10[(45)^2 - y^2]$$

$$y^2 - 9P_0 y - (45)^2 = 0$$

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Ex! A thin tube of uniform cross-section is sealed at both the ends, it lies horizontally. the middle 5cm containing mercury & the 2 equivalents containing air at pressure P_0. the tube is held at 60° with the vertical as shown. Calculate pressure P_0 in cm of Hg.





$$L + 5 + L = 46 + 5 + 44.5$$

$$2L = 90.5$$

$$\boxed{L = 45.25\text{cm}}$$

∵ Hg is in eqm,

$$P_B A + mg\cos 60^\circ = P_A A$$

$$P_B A + hA\,sg \times \frac{1}{2} = P_A A$$

$$P_B + \frac{5}{2} sg = P_A$$

$$P_A - P_B = 2.5 \text{ cm of Hg}$$

from boyle's law,

$$P_0(45.25) = P_A(44.5)$$

$$P_A = P_0 \frac{(45.25)}{44.5}$$

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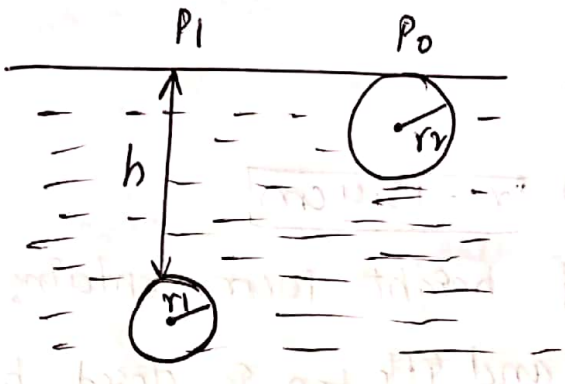
$$P_0(45.25) = P_B(46)$$

$$P_B = \frac{P_0(45.25)}{46}$$

$$\frac{P_0(45.25)}{44.5} - \frac{P_0(45.25)}{46} = 2.5$$

$\Rightarrow$ by solving, $P_0 = 76.4$ cm of Hg.

Air Bubble in a Lake:



$P_1 V_1 = P_2 V_2$

temp of lake is const

$$\left(P_0 + h\rho g + \frac{2T}{R_1}\right) V_1 = \left(P_0 + \frac{2T}{R_2}\right) V_2$$

$P_0 = 76$ cm of Hg

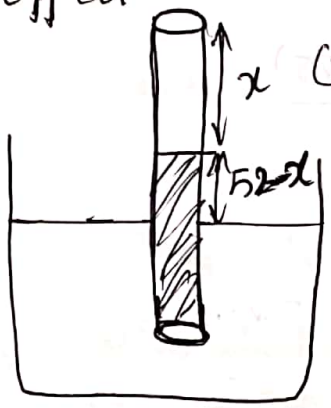
$$= \frac{76}{100} \times 13.6 \times 10^3 \times 9.8$$

$= 10.28$ m of $H_2O = 10.28 \times 10^3 \times 9.8$.

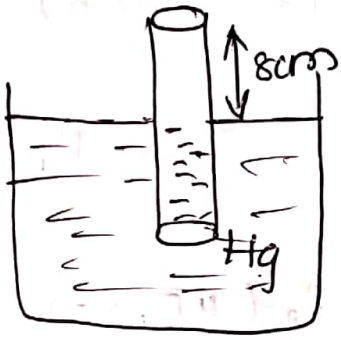
Ex: An open glass tube is immersed in Hg in such a way that the length 8cm extends above the Hg column. now the open end of tube is closed by finger and raised further by 44cm. what will be the length of air column above the Hg in the tube the atm pressure

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to be 76 cm of Hg. Neglect the capillary effect.



(8+44)cm
=52cm



P' + (52-x) cm of Hg = P_0

$P' = P_0 - (52-x)$

$= 76 - 52 + x$

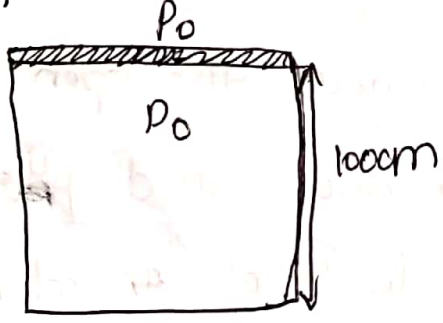
$= 24 + x$ cm of Hg.

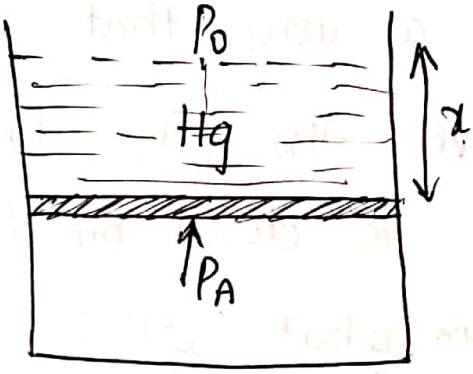
from boyle's law,

$PL = \text{const}$

$\therefore 76 \times 8 = (24+x)x \Rightarrow \boxed{x = 15.4 \text{ cm}}$

Ex: A vertical cylinder of height 100cm contains air at a const temp and its top is closed by a frictionless piston at atmospheric pressure. If Hg is slowly poured on the piston due to its weight air is compressed. find the max height of the Hg column which can be put on the piston.





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since piston is in eqm,

$$P_0 A + x\rho g A = P_A A$$

$$P_A = P_0 + x\rho g$$

$$P_A = (H + b) \text{ cm of Hg}$$

Boyle's Law,

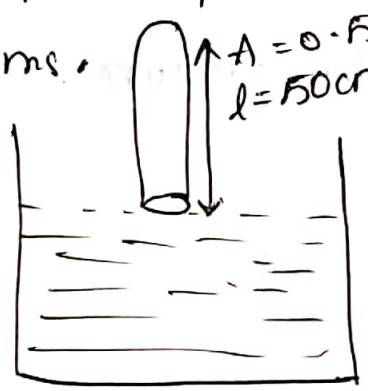
$$P_0 (100) = P(100 - x)$$

$$76(100) = (76 + x)(100 - x)$$

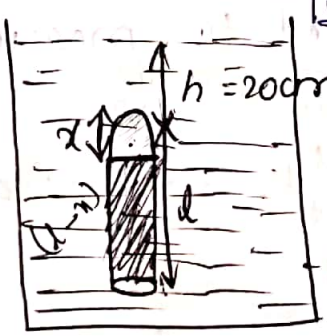
$$\boxed{x = 24 \text{ cm}}$$

Ex:

A glass tube of length 50cm and cross-section $A = 0.5 cm^2$ is sealed at one end and submerged into water as shown. What force should be applied to hold the tube under the water if the distance from the surface of the water to the closed end is $h = 20cm$ and the atmospheric pressure is 10^5 Pa. The wt of the tube is 15 gms.



0.084N



$V_d \rho g$
↙
Ax.

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From boyle's law,

$$P_1 = \text{const}$$

$$P_0 l = [P_0 + h + l - (l - x)]\, x \Rightarrow P_0 l = [P_0 + h + x]\, x$$

$$10^5 \times \frac{50}{100} = \left[10^5 + \frac{20}{100} \times 10^3 \times 10 + x \times 10^3 \times 10\right] x$$

$$5 = (10 + 0.2 + x)\, x$$

$$5 = 10.2x + x^2$$

$$x^2 + 10.2x - 5 = 0$$

$$\boxed{x = 0.468\,m}$$

now, Net force on the tube,

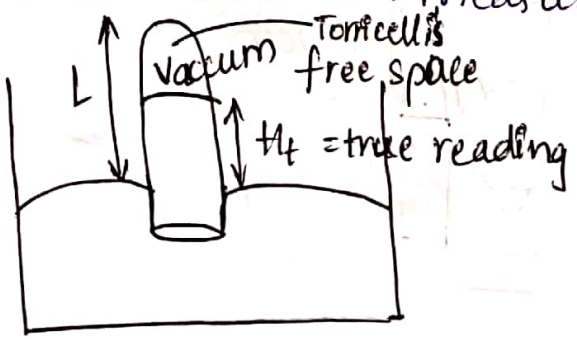
$$F_{net} = F_B - mg$$

$$= A x \rho_w g - mg$$

force required to hold the tube = F_{net} but in opp direction.

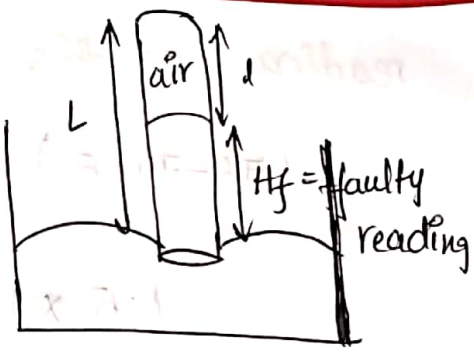
Faulty Barometer

* instrument used to measure the pressure.



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If some air trapped in the tube above the mercury.



$P_{air} + H_f\, sg = P_0$

$P_{air} + H_f\, sg = H_t\, sg$

$P_{air} = (H_t - H_f)\, sg$ cm of Hg.

$$\boxed{P = H_t - H_f}$$

Length of air, $d = L - H_f$

Pressure of air, $P = H_t - H_f$

now, from Boyle's law,

$Pd = \text{const}$

$$\boxed{(H_t - H_f)(L - H_f) = \text{const}}$$

In general,

$$\boxed{(H_1 - H_1')(L - H_1') = (H_2 - H_2')(L - H_2')}$$

Here H_1, H_2 are true readings.

H_1', H_2' are fault readings.

Ex: A faulty Barometer tube is 90cm long and it contains some air above Hg. The reading is 74.5cm and the true atm pressure is 76cm. what will be true atm pressure if the

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reading on this barometer is 74cm.

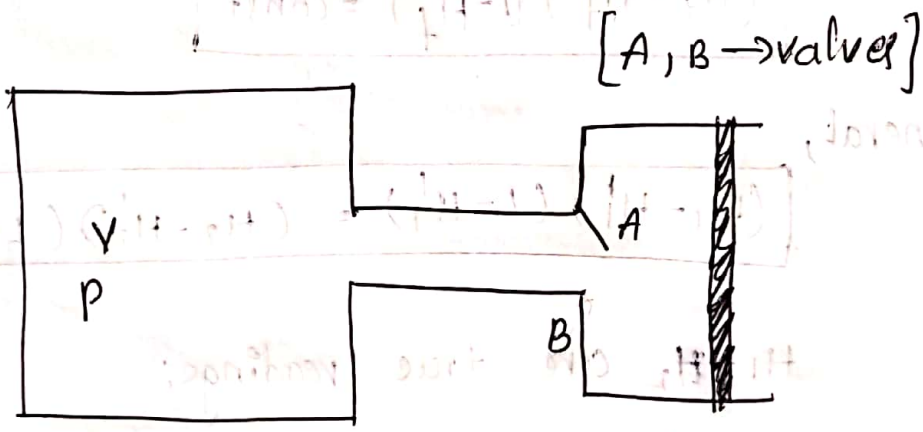
$$(76-74.5)(90-74.5) = (H_2-74)(90-74)$$

$$1.5 \times 15.5 = (H_2-74) \times 16$$

$$H_2 - 74 = \frac{15 \times 155}{1600}$$

$$\boxed{H_2 = 75.45}$$

Ex:
A gas is enclosed in a vessel of volume 'V' at a pressure 'P' it is being pumped out of the vessel by means of piston pump with a stroke volume ΔV. what is the final volume of the vessel after n strokes. assume temp remains const.



from bogle's law,

$$PV = P_1(V+\Delta V)$$

after first stroke

$$P_1 = \frac{PV}{V+\Delta V}$$

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for next stroke

$$P_1 V = P_2 (V + \Delta V)$$

after second stroke

$$P_2 = \frac{P_1 V}{V + \Delta V} = \frac{P V^2}{(V + \Delta V)^2}$$

After nth stroke,

$$P_n = \frac{P V^n}{(V + \Delta V)^n}$$

$$\frac{P_n}{P} = \left(\frac{V}{V + \Delta V}\right)^n$$

$$\ln \frac{P_n}{P} = n \ln \left(\frac{V}{V + \Delta V}\right)$$

$$\boxed{n = \frac{\ln (P_n / P)}{\ln (V / V + \Delta V)}}$$

① Here escaped air mass is not const during each stroke.

② from ideal gas eqn,

$$PV = \frac{m}{M} RT$$

$$P_{n-1} V = \frac{m_{n-1}}{M} RT$$

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$$P_n V = \frac{m_n}{M} RT$$

$$m_{n-1} - m_n = (P_{n-1} - P_n) \frac{VM}{RT}$$

mass of air escaped during n^{th} stroke.

multiplying with $\frac{V}{m_n m_{n-1}}$

$$\frac{V}{m_n} - \frac{V}{m_{n-1}} = (P_{n-1} - P_n) \left[\frac{VM}{RT} \times \frac{V}{m_n m_{n-1}} \right]$$

Again we have,

$$P_{n-1}(V) = P_n(V + \Delta V)$$

$$(P_{n-1} - P_n) V = P_n (\Delta V)$$

$$\frac{V}{m_n} - \frac{V}{m_{n-1}} = \left(\frac{P_n \Delta V}{V}\right)\left(\frac{VM}{RT}\right)\left(\frac{V}{m_n m_{n-1}}\right)$$

$$\frac{P_n VM}{RT} = m_n$$

$$\frac{V}{m_n} - \frac{V}{m_{n-1}} = m_n \frac{\Delta V}{V} \times \frac{V}{m_n \times m_{n-1}}$$

$$\frac{\frac{V}{m_n} - \frac{V}{m_{n-1}}}{\frac{V}{m_{n-1}}} = \frac{\Delta V}{V} = \text{const}$$

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(c) fractional change in specific volume of gas = const.

(2) Charle's Law at const pressure:

1) Vol coeff of gas (α):

$$\alpha = \frac{1}{V_0}\frac{\Delta V}{\Delta t} \ /^\circ C \text{ or } /K$$

$$\alpha = \frac{1}{273.15} \ /^\circ C$$

* At const pressure the volume of given mass of gas varies linearly with temp as,

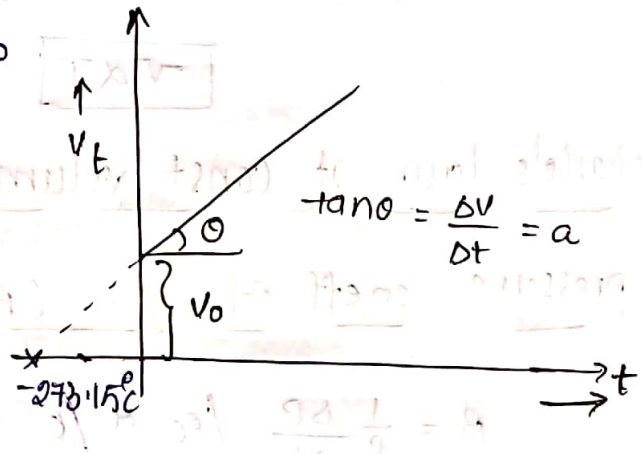
$$\boxed{V_t = at + b} \qquad t \rightarrow {}^\circ C$$

At $t = 0^\circ C$, $V_t = V_0$

$$V_t = \frac{\Delta V}{\Delta t} \times t + V_0$$

$$V_t = \alpha V_0 t + V_0$$

$$\boxed{V_t = V_0(1 + \alpha t)}$$



$$V_{t_1} = V_0(1 + \alpha t_1)$$

$$V_{t_2} = V_0(1 + \alpha t_2)$$

$$\Rightarrow \boxed{\alpha = \frac{V_{t_2} - V_{t_1}}{V_{t_1} t_2 - V_{t_2} t_1} \ /^\circ C}$$

Again,

** $V_t = V_0\left(1 + \frac{t}{273.15}\right)$

$$V_t = V_0\left(\frac{t + 273.15}{273.15}\right) = \frac{V_0 T}{273.15} \Rightarrow \boxed{V \propto T}$$

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$V_t = 0$ when $t = -273.15\,^\circ C$

(or) $T = +273.15$

$$\boxed{T = 0\,K}$$

* Vol of gas is directly proportional to absolute temp at const pressure & for a given mass of gas.

i.e, $$\boxed{\frac{V_1}{V_2} = \frac{T_1}{T_2}}$$

II-Method:

From kinetic theory,

$$PV = \frac{1}{3} m V_{rms}^2$$

here $m, P \rightarrow$ const

$$V \propto V_{rms}^2$$

$$\boxed{V \propto T}$$

③ Charle's law at const volume (Gay-lussac's law):

pressure coeff of gas (β):

$$\beta = \frac{1}{P_0}\frac{\Delta P}{\Delta t} \quad /^\circ C \text{ or } /K$$

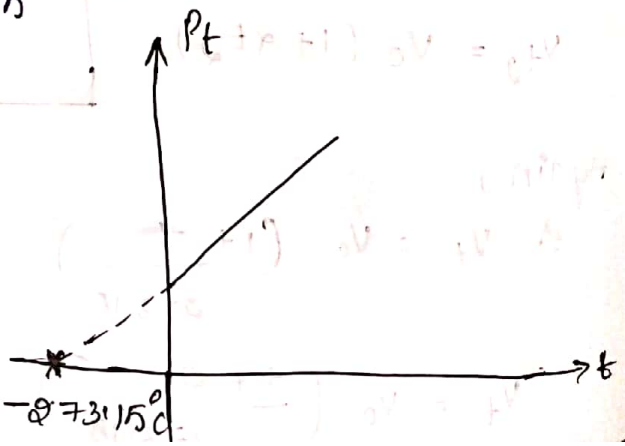
Clearly $\alpha = \beta = \frac{1}{273.15} \quad /^\circ C$

Again, $P_t = at + b$

$$P_t = P_0(1 + \beta t)$$

$$P_1 = P_0(1 + \beta t_1)$$

$$P_2 = P_0(1 + \beta t_2)$$



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$\Rightarrow$ $\boxed{\beta = \frac{P_2 - P_1}{P_1 t_2 - P_2 t_1} \text{ /°C}}$

Finally,

$P_t = 0$ at $t = -273.15°C$

(or) $\boxed{T = 0K}$

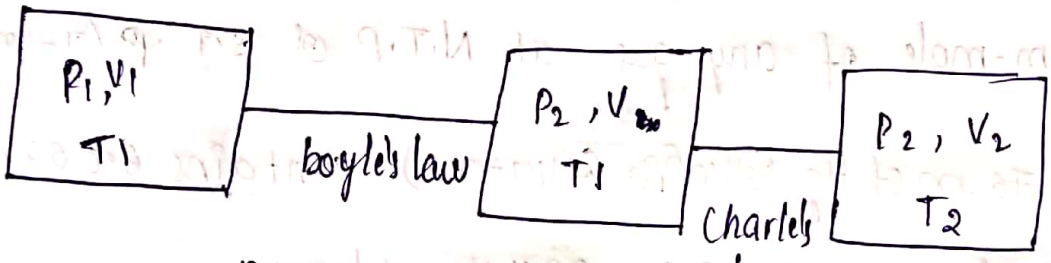
& $\boxed{P \propto T}$

$$\boxed{\frac{P_1}{P_2} = \frac{T_1}{T_2}}$$

Ideal Gas Eqn :

└ gas that obeys boyle's, charle's law at const temp & pressure.

① Consider 1 gm of Ideal gas



$P_1 V_1 = P_2 V$

$\Rightarrow V = \frac{P_1 V_1}{P_2}$ — ①

$\frac{V}{T_1} = \frac{V_2}{T_2}$

$V = \frac{V_2 T_1}{T_2}$ — ②

From eqn ① & ②,

$$\boxed{\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}}$$

* The ratio $\frac{PV}{T}$ per unit mass of gas is called

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"Specific gas constant (r)".

i.e, $\frac{PV}{T} = r$

for m-gms of gas

$$\frac{PV}{mT} = r$$

$$\boxed{PV = mrT}$$

* units for r ⟶ $J\,Kg^{-1}K^{-1}$

$cal\,gm^{-1}\,{}^{o}C^{-}$

* r is different for different gases.

Here $\boxed{r = \frac{PV}{mT} = \frac{P}{\rho T}}$

* If mass of gas is taken in 'gm-mole', then 1gm-mole of any gas at N.T.P or S.T.P ($T = 273K$, $P = 76cm$ of Hg (or) $10^5 Pa$ (or) 1 atm) contains 6.023×10^{23} molecules & occupied 22.4 lts volume.

↳ $22.4 \times 10^{-3} m^3$

* The ratio $\frac{PV}{T}$ per gm-mole is called "Universal gas const (R)".

i.e, $\frac{PV}{T} = R$

for n-moles of gas

$$\boxed{PV = nRT}$$

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* R ⟶ 8.314 J $mol^{-1} K^{-1}$
& cal $mol^{-1} K^{-1}$

*R is same for all gases.

Dalton's Law of partial Pressure:

$P = P_1 + P_2 + ----$

resultant pressure is equal to sum of partial pressures.

$PV = \frac{1}{3} m v_{rms}^2$

$PV = nRT$

$P = \frac{nRT}{V}$

Graham's Law of diffusion:

rate of diffusion $r \propto v_{rms} \propto \frac{1}{\sqrt{\rho}}$

$$\frac{r_1}{r_2} = \frac{v_{rms_1}}{v_{rms_2}} = \sqrt{\frac{\rho_2}{\rho_1}}$$

$$v_{rms} = \sqrt{\frac{3P}{\rho}}$$

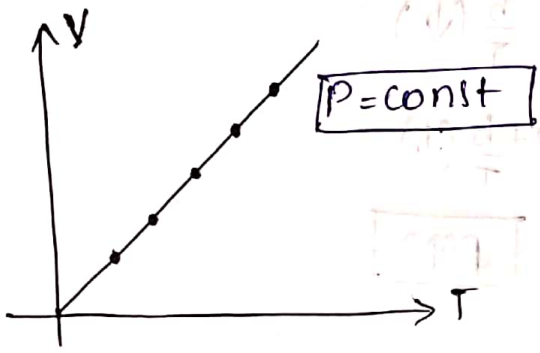
Avagadro's Law:

$PV = NKT$

$P, V, T \longrightarrow$ const $\Rightarrow$ N = const

Graphs:

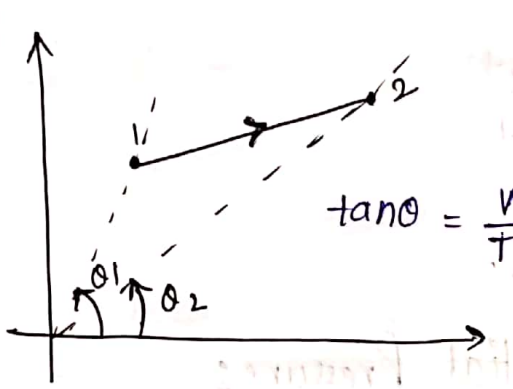
①



P = const

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(2)



$\tan\theta = \frac{V}{T} = \frac{nR}{P}$

$\theta_1 > \theta_2$

$\tan\theta_1 > \tan\theta_2$

$\frac{nR}{P_1} > \frac{nR}{P_2}$

$\boxed{P_2 > P_1}$

i.e, pressure is increasing continuously.

M-II:

$V = aT + b$

we have, $PV = nRT$

$P = \frac{nRT}{V}$

$P = \frac{nRT}{aT+b}$

$P = \frac{nR}{a + \frac{b}{T}}$

As $T(\uparrow) \Rightarrow \frac{b}{T}(\downarrow)$

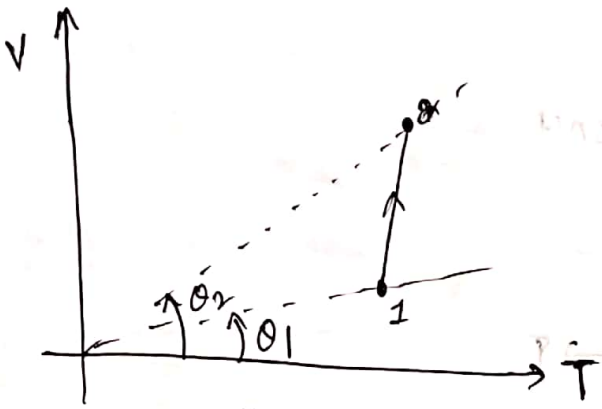
$\Rightarrow a + \frac{b}{T}(\downarrow)$

$\Rightarrow \boxed{P(\uparrow)}$



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(3)



$\theta_1 < \theta_2$

$\tan\theta_1 < \tan\theta_2$

$\frac{nR}{P_1} < \frac{nR}{P_2}$

$\boxed{P_2 < P_1}$

⇒ pressure is decreasing continuously.

N-II : $V = aT - b$

we have, $PV = nRT$

$$P = \frac{nRT}{V}$$

$$P = \frac{nRT}{aT - b}$$

$$P = \frac{nR}{a - \frac{b}{T}}$$

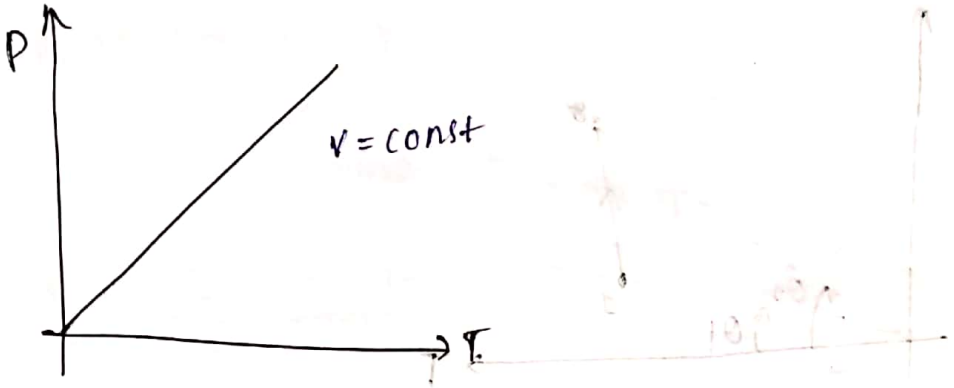
As $T(\uparrow) \Rightarrow \frac{b}{T}(\downarrow)$

$\Rightarrow a - \frac{b}{T}(\uparrow)$

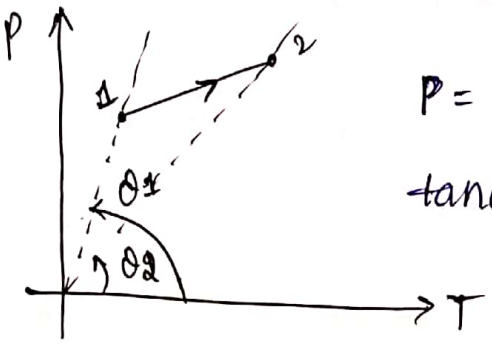
$\Rightarrow \boxed{P \downarrow}$

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④



⑤



$P = \frac{nR}{V} T$

$\tan\theta = \frac{nR}{V}$

$\theta_1 > \theta_2$

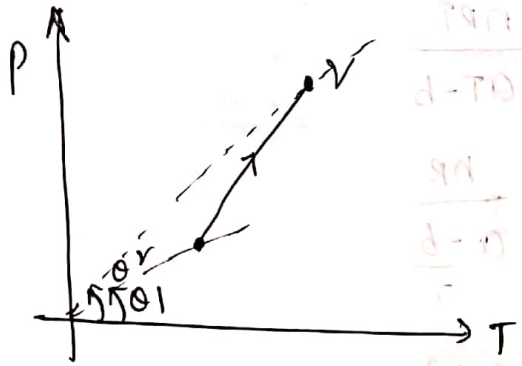
$\tan\theta_1 > \tan\theta_2$

$\frac{nR}{V_1} > \frac{nR}{V_2}$

$\boxed{V_2 > V_1}$

i.e, V ↑ continuously

⑥



$\theta_2 > \theta_1$

$\Rightarrow \boxed{V_1 > V_2}$

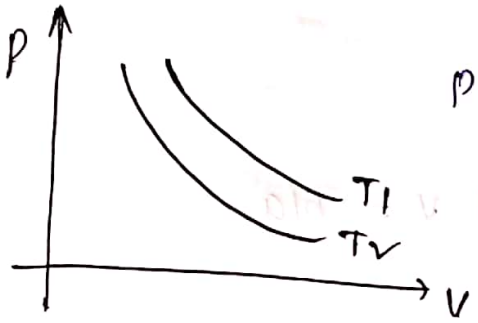
i.e, V↓ continuously.

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(7)



(8)

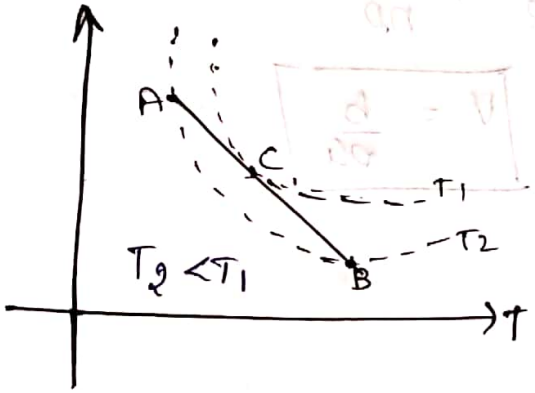


$PV = nRT$

$PV \propto T$

$\boxed{T_1 > T_2}$

(9)



*temp of gas first increase and then decreases.

* gas is first heated & then cooled.

To Find Max temp:

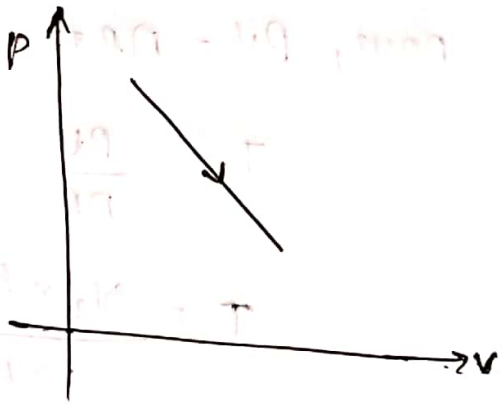
Eqn of st. line,

$P = -aV + b$

we have,

$PV = nRT$

$(-aV + b)V = nRT$

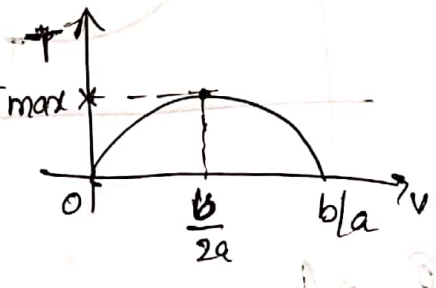


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$$-aV^2 + bV = nRT$$

$$T = \frac{-aV^2}{nR} + \frac{bV}{nR}$$

$$\boxed{y = ax - bx^2}$$



$T = 0$

$$\Rightarrow -\left(\frac{aV}{nR} + b\right)V = 0$$

$$\Rightarrow V = 0 \text{ and } V = b/a$$

$T = T_{max}$

$$\frac{dT}{dV} = 0 \Rightarrow -\frac{2aV}{nR} + \frac{b}{nR} = 0$$

$$\boxed{V = \frac{b}{2a}}$$

At this volume,

$$P = -a\left(\frac{b}{2a}\right) + b$$

$$P = -\frac{b}{2} + b$$

$$\boxed{P = b/2}$$

now, $PV = nRT$

$$T = \frac{PV}{nR}$$

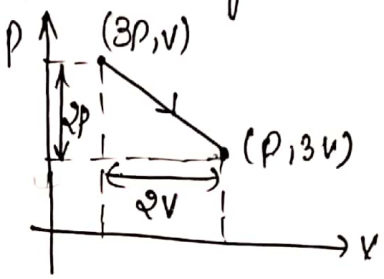
$$T = \frac{b/2 \times b/2a}{nR} = \frac{b^2}{4anR}$$

(or) $$T_{max} = -\frac{a}{nR}\left(\frac{b}{2a}\right)^2 + \frac{b}{nR}\left(\frac{b}{2a}\right) = \frac{b^2}{4anR}$$

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$$T_{max} = \frac{b^2}{4aR}$$

→ Find the maximum temp of the gas.



$P = -aV + b$

$3P = -aV + b$ — ①

$P = -3aV + b$ — ②

① - ②

$2P = +2aV$

$P = aV$

$$a = \frac{P}{V}$$

② ⇒ $P = -3a \times V + b$

$P = -3 \times \frac{P}{V} \times V + b$

$4P = b$

$(\because T_{max} = \frac{b^2}{4aR})$

$T_{max} = \frac{16P^2}{4 \times \frac{P}{V} nR}$

$$T_{max} = \frac{4PV}{nR}$$

Short-cut:

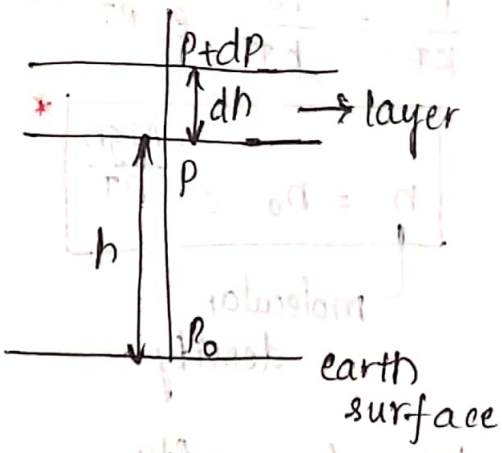
only when $\Delta P = \Delta V$

$$T_{max} = \frac{P_{avg} \times V_{avg}}{nR}$$

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Variation of pressure with altitude (h):

Here Temperature and molecular wt of composition of atmosphere assumed to be same.



In eqm,

$$(P+dP)A + (dm)g = PA$$

$$dPA + dmg = 0$$

$$dPA = -dmg$$

$$dPA = -A\,dh\,\rho g$$

$$dP = -dh\rho g$$

we have,

$$PV = nRT$$

$$PV = \frac{m}{M}RT$$

$$P = \frac{m}{V}RT$$

$$\boxed{\frac{PM}{RT} = \rho}$$

$$\boxed{dP = -\frac{PM}{RT}g\,dh} \quad *$$

$$\int_{P_0}^{P}\frac{dP}{P} = -\frac{Mg}{RT}\int_0^h dh$$

$$\ln(P/P_0) = -\frac{Mgh}{RT} \quad *$$

$$\boxed{P = P_0 e^{-\frac{Mgh}{RT}}} \quad *$$

i.e, pressure of gas decreases exponentially with height h.

& $\frac{\rho RT}{M} = \frac{\rho_0 RT}{M} e^{-\frac{Mgh}{RT}}$

$$\boxed{\rho = \rho_0 e^{-\frac{Mgh}{RT}}} \quad *$$

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density of air decreases exponentially with height (h).

Again,

$$\frac{P}{KT} = \frac{P_0}{KT} e^{\frac{-Mgh}{RT}}$$

$$\boxed{n = n_0 e^{\frac{-Mgh}{RT}}} \quad *$$

n → molecular density

Centre of gravity of atmosphere :

$$h_{CG} = \frac{\int_0^\infty (dm) gh}{\int_0^\infty (dm) g} \qquad \text{where } dm = A\,dh\,\rho$$

$$= A\,dh\,\rho_0 e^{\frac{-Mgh}{RT}}$$

$$\therefore h_{CG} = \frac{\int_0^\infty A g \rho_0 e^{\frac{-Mgh}{RT}} \cdot h \cdot dh}{\int_0^\infty A g \rho_0 e^{\frac{-Mgh}{RT}} \cdot dh} = \frac{\int_0^\infty e^{\frac{-Mgh}{RT}} \cdot h \cdot dh}{\int_0^\infty e^{\frac{-Mgh}{RT}} \cdot dh}$$

$$\boxed{h_{CG} = \frac{RT}{Mg}} \quad *$$

Ex: In the lower atm the temp decreases with altitude. given that the temp variation is linear $T = T_0(1 + \epsilon y)$ where the reference level is at the surface of the earth. Find an expression for the variation of pressure with altitude.

Sol: $T = T_0(1 + \epsilon y)$

In eqm,

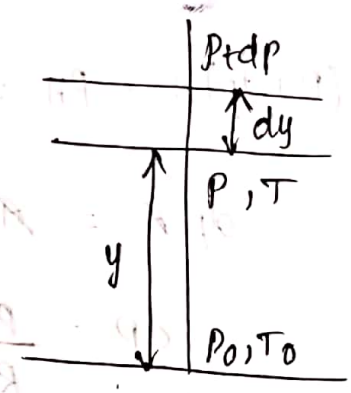
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$$(P + dp)A + dmg = PA$$

$$(dp)A = -(dm)g$$

$$dp\,A = -A\,dy\,\rho g$$

$$dp = -\frac{PMg}{RT}\,dy$$



$$\frac{dp}{P} = -\frac{Mg}{RT_0(1-cy)}\,dy$$

$$\int_{P_0}^{P}\frac{dp}{P} = \frac{-Mg}{RT_0}\int_0^y \frac{dy}{(1-cy)}$$

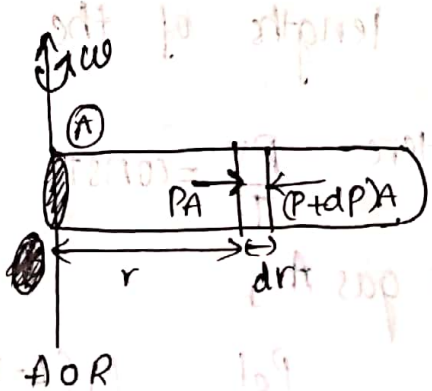
$$\ln(P/P_0) = \frac{+Mg}{RcT_0}\left[\ln(1-cy)\right]_0^y$$

$$\ln(P/P_0) = \frac{Mg}{RcT_0}\ln(1-cy)$$

$$\ln\frac{P}{P_0} = \ln(1-cy)^{\frac{Mg}{RcT_0}}$$

$$\boxed{P = P_0(1-cy)^{\frac{Mg}{RcT_0}}}$$

Ex: Find variation of pressure as a function of distance (r) from axis of rotation.



Sol: Net force towards the centre,

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$$(P+dp)A - PA = dm(r\omega^2)$$

$$dp\,A = A\,dr\,\rho\,r\omega^2$$

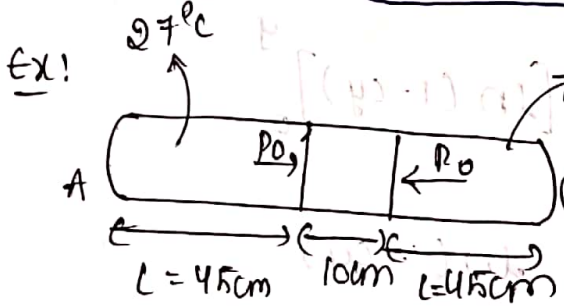
$$dp = \frac{PM}{RT}\omega^2 r\,dr$$

$$\int_{P_0}^{P}\frac{dP}{P} = \frac{M\omega^2}{RT}\int_0^r r\,dr$$

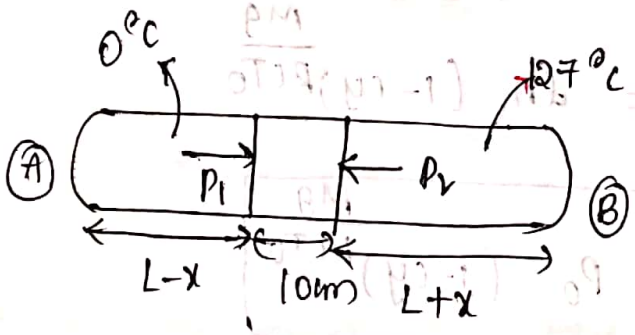
$$\ln(P/P_0) = \frac{M\omega^2}{RT}r^2$$

$$\boxed{P = P_0 \cdot e^{\frac{M\omega^2 r^2}{RT}}}$$

Ex!



—initial



—final

$x = 8.49\text{cm}$

$P_1 = P_2 = 85.25\text{ cm of Hg}$

Find lengths of the air columns & pressures.

Sol:

Here - $\frac{PV}{T} = \text{const}$

For gas A,

$$\frac{P_0 L}{300} = \frac{P_1(L-x)}{273} \quad \text{(1)}$$

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For Gas B,

$$\frac{P_0 L}{300} = \frac{P_2 (L+x)}{400} \quad -(2)$$

In eqm, $P_1 A = P_2 A$

$$\boxed{P_1 = P_2} \quad -(3)$$

(1)=(2)

$$\frac{L-x}{273} = \frac{L+x}{400} \Rightarrow 400L - 400x = 273L + 273x$$

$$127L = 673x$$

$$x = \frac{127 \times 45}{673}$$

$$\boxed{x = 8.492\ cm}$$

$$\frac{P_0 \times 45 \times 273}{300 \times 36.508} = P_1$$

$$P_1 = \frac{76 \times 45 \times 273}{300 \times 36.508} = 85.247\ cm\ of\ Hg$$

Ex: A barometer tube 90 cm long contained some air above mercury. The reading is 74.5 cm when the true pressure is 76 cm & temp is 15°C. If the reading is to be observed 75.8 cm on a day when the temp is 5°C, what is the true pressure.

Sol: pressure of air = True pressure − Faulty reading

length of air = Tube length − reading

At 15°C,

$T_1 = 15 + 273 = 288\ K$

$P_1 = 76 - 74.5 = 1.5\ cm$

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$l_1 = 90 - 74.5 = 15.5\,cm$, $V_1 = A l_1$

At 5°C

$T_2 = 5 + 273 = 278\,K$

$P_2 = ?$

$l_2 = 90 - 75.8 = 14.2\,cm$

$V_2 = A l_2$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1.5 \times 15.5}{288} = \frac{P_2 \times 14.2}{278}$$

$$P_2 = \frac{6463.5}{288 \times 14.2} = \frac{22.4}{14.2} = 1.58$$

True pressure $= 75.8 + 1.58$

$= 77.38$

Ex: An air bubble comes from bottom to the surface of a lake of depth 0.5 m. the surface temperature of lake is 4°C. The diameter of the bubble at the bottom & at the surface are 3.6mm & 4mm respectively. Find the temperature of the lake at the bottom.

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$$\frac{P_s V_s}{T_s} = \frac{P_b V_b}{T_b}$$

$$\frac{10^5 \times r_s^3}{313} = \frac{(10^5 + 0.25 \times 10^5)\, r_b^3}{T_b}$$

$$T_b = 1.25 \times 313 \times \left(\frac{r_b}{r_s}\right)^3$$

$$= 1.25 \times 313 \times (0.9)^3$$

$$T_b = 1.25 \times 313 \times 0.729$$

$$T_b = 285.82\,K$$

$$\boxed{T_b = 12.82\,^\circ C}$$

Ex: Find min attainable pressure of 1 mol of an ideal gas during it's expansion if temp & vol are related as $T = T_0 + \alpha V^2$. where $T_0, \alpha \longrightarrow$ +ve constants

Given $T = T_0 + \alpha V^2$

we have, $PV = nRT$, $n = 1$ mole

$$P = \frac{RT}{V}$$

$$P = \frac{R(T_0 + \alpha V^2)}{V}$$

$$P = \frac{RT_0}{V} + R\alpha V$$

for $P = P_{min}$,

$$\frac{dP}{dV} = 0$$

$$-\frac{RT_0}{V^2} + R\alpha = 0$$

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$$\frac{T_0}{V^2} = \alpha$$

$$\boxed{V = \sqrt{\frac{T_0}{\alpha}}}$$

and $T = T_0 + \alpha \left(\frac{T_0}{\alpha}\right)$

$$\boxed{T = 2T_0}$$

So at $V = \sqrt{\frac{T_0}{\alpha}}$, $T = T_0$

pressure is minimum.

$$P_{min} = RT_0\sqrt{\frac{\alpha}{T_0}} + R\alpha\sqrt{\frac{T_0}{\alpha}}$$

$$\boxed{P_{min} = 2R\sqrt{T_0\alpha}}$$

Ex: The density of 'H' is 0.08696 gm/Lt at S.T.P. Find gas const of Hydrogen.

Sol: given,

$$\rho = 0.08696 \text{ gm/Lt}$$

i.e) mass of 1 Lt volume of H is $m = 0.08696$ gm

i.e $V = 1$ lt

$= 1000 \text{ cm}^3$

At S.T.P, $P = 76$ cm of Hg

$= 76 \times 13.6 \times 980$ dyne/cm^2

$T = 273$ K

we have $PV = mRT$

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$$R = \frac{PV}{mT} = \frac{76 \times 13.6 \times 980 \times 1000}{0.0896 \times 273}$$

$$= 4.267 \times 10^7 \text{ erg gm}^{-1} \text{K}^{-1}$$

$$= 4.267 \text{ J gm}^{-1} \text{K}^{-1}$$

$$= 4267 \text{ J kg}^{-1} \text{K}^{-1}$$

Important pt's

$$PV = mRT \quad -(1)$$

and $PV = nRT$

$$PV = \frac{m}{M} RT \quad -(2)$$

From eqs (1) & (2)

$$mRT = \frac{m}{M} RT$$

$$\boxed{r = \frac{R}{M}}$$

Ex: The mass of a lit of a gas is 1.562 gm at 0°C under a pressure of 76cm of Hg. The temp is increased to 25°C and the pressure of 78 cm of Hg. what is the mass of 1 Lt of gas under these condns.

we have,

$$PV = nRT$$

$$PV = \frac{m}{M} RT$$

$$\frac{PV}{mT} = \text{const}$$

$$\frac{P_1 V_1}{m_1 T_1} = \frac{P_2 V_2}{m_2 T_2}$$

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$$\frac{7638}{1.562 \times 273} = \frac{7839}{m \times 523}$$

$$\frac{1}{m} = \frac{38 \times 523}{39 \times 1.562 \times 273}$$

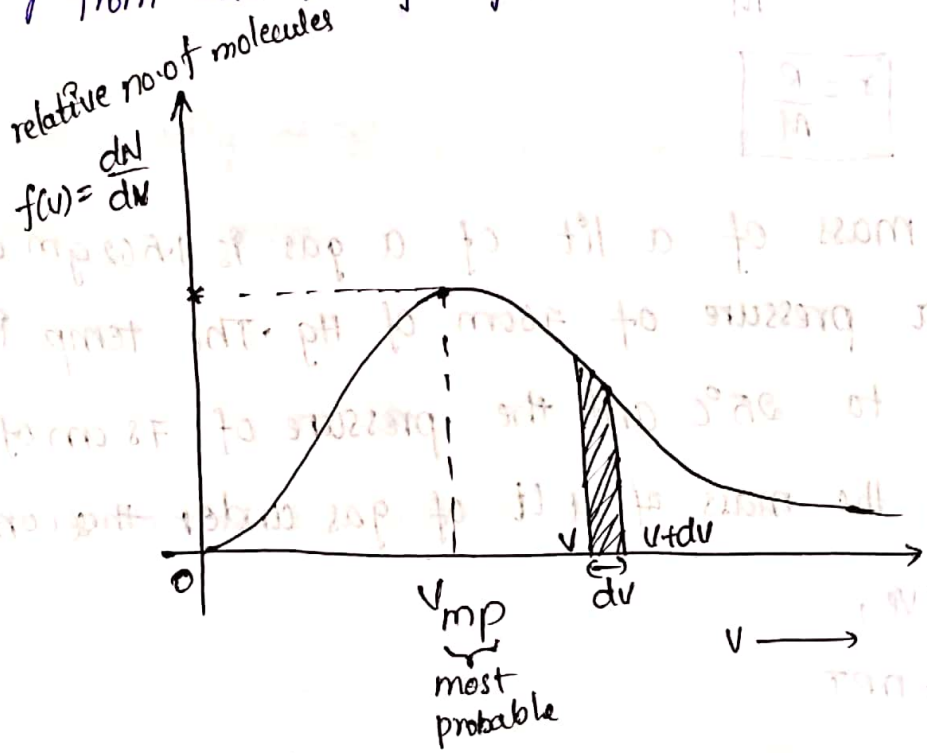
$$\frac{1}{m} = \frac{19874}{16630.614}$$

$$\boxed{m = 0.836 \text{ gms}}$$

Maxwell's Speed distribution Law :

At a particular Temperature, all the particles of the gas are not moving with same speed. Their speeds are ranging from zero to infinity.

①



Area of elementary strip

$= f(v) \cdot dv$

$= \frac{dN}{dv} \cdot dv$

$= dN$ = no. of molecules having speeds in between v & $v+dv$.

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Total area of region under curve & X-axis,

$$N = \int_0^\infty dN = \text{Total no. of particles of gas.}$$

From kinetic theory,

$$dN = f(v)\cdot dv = 4\pi N v^2 \left(\frac{m}{2\pi RT}\right)^{3/2} e^{-\frac{mv^2}{2RT}}\, dv$$

Different Speeds of molecules:

① **RMS speed:**

$$v_{rms} = \sqrt{\frac{N_1 v_1^2 + N_2 v_2^2 + \cdots}{N_1 + N_2 + \cdots}} \quad *$$

(or) $$* \; v_{rms} = \sqrt{\frac{\int_{v_p}^{v_t} v^2 \cdot dN}{\int_{N_1}^{\infty} dN}}$$

$$= \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m_0}} = \sqrt{\frac{3P}{\rho}}$$

here, $v_{rms} \propto \sqrt{T}$ [m = const]

$v_{rms} \propto \sqrt{M}$ (T = const)

and, if pressure increase then density also increases (at const temp)

$\Rightarrow$ **v_{rms} remains same** *

we have $\rho = \frac{PM}{RT} \Rightarrow P \propto \rho$

* Translational kinetic energy:

$$K.E = \frac{1}{2} m_0 v_{rms}^2 = \frac{1}{2} m_0 \frac{3kT}{m_0} = \frac{3}{2} kT \Rightarrow \boxed{K.E = \frac{3}{2} kT}$$

$\boxed{K.E \propto T}$ * $K.E = 0$ at absolute Temp (0K)

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* K.E of m-gms of gas,

$$K.E = \frac{1}{2} m v_{rms}^2 = \frac{1}{2} \times m \times \frac{3P}{\rho} = \frac{3}{2} PV$$

$$\boxed{K.E = \frac{3}{2} PV}$$

$$\frac{K.E}{V} = \frac{3}{2} P \Rightarrow \boxed{\epsilon = 3/2\, P}$$

ϵ → kinetic energy per unit volume of gas.

② Avg Speed (v_{avg})

Avg velocity,

$$\bar{V}_{avg} = \frac{\bar{v}_1 + \bar{v}_2 + - - - - \bar{v}_N}{N}$$

Since molecules are moving at random,

$$\boxed{\bar{V}_{avg} = 0}$$

Avg speed,

$$v_{avg} = \frac{|\bar{v}_1| + |\bar{v}_2| + - - - + |\bar{v}_N|}{N}$$

$$= \frac{v_1 + v_2 + v_3 + - - - + v_N}{N}$$

In general,

$$\boxed{v_{avg} = \frac{N_1 v_1 + N_2 v_2 + - - - - - n}{N_1 + N_2 + - - - -}} \quad *$$

From kinetic theory,

$$* \; v_{avg} = \frac{\int_{v_i}^{v_f} v \cdot dN}{\int_{v_0}^{v_f} dN}$$

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$$V_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8kT}{\pi m_0}} = \sqrt{\frac{8P}{\pi \rho}}$$

③ Most probable Speed (V_{mp})

It is the speed possessed by max no. of molecules of the gas.

For $V = V_{mp}$,

$$\therefore \frac{d}{dv}(f(v)) = 0$$

$$\Rightarrow \boxed{V_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2kT}{m_0}} = \sqrt{\frac{2P}{\rho}}}$$

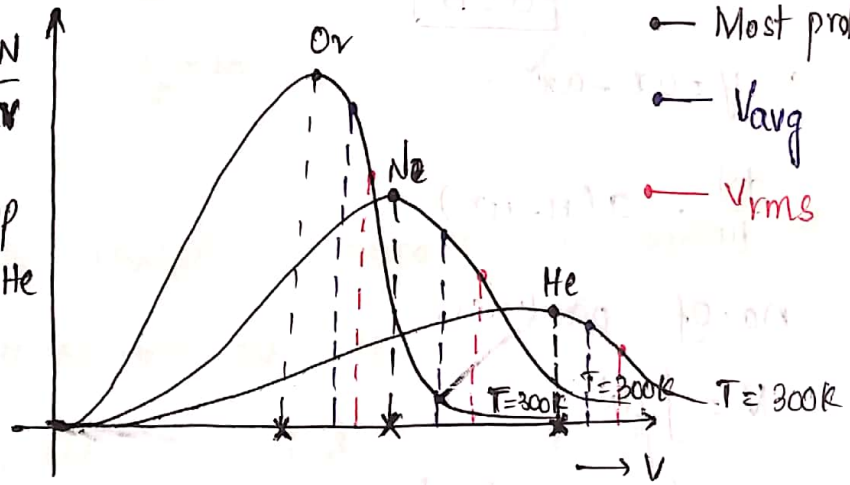
Clearly $V_{rms} > V_{avg} > V_{mp}$

& $\boxed{V_{rms} : V_{avg} : V_{mp} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2}}$ *

Graphs:

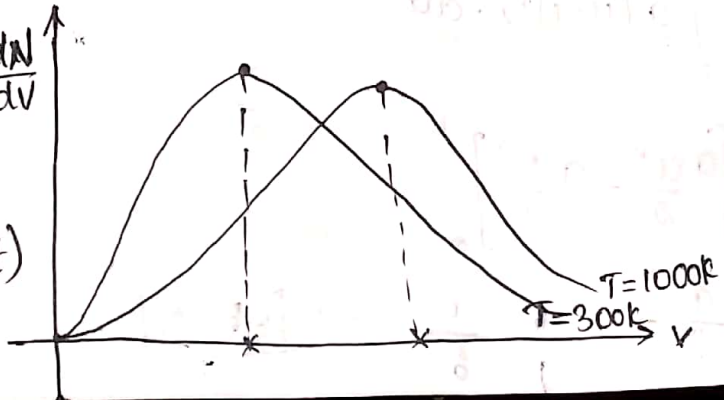
1) $\frac{dN}{dv}$

at same temp for O_2, Ne, He



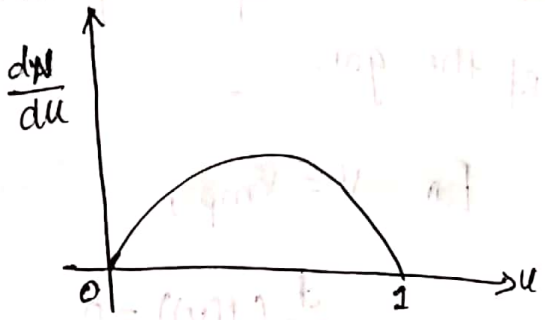
2) $\frac{dN}{dV}$

for same gas (M = const)



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Ex: The speed distribution of molecules in a sample of gas is as shown in the figure. The graph b/w $\frac{dN}{du}$ Vs u is a parabola & Total no of molecules in the sample is N_0. Calculate



i) rms speed

ii) V_{avg}

iii) V_{mp}

iv) Total translational K.E of molecules if the mass of the sample is 10gms.

Eqn of parabola, $y = ax - bx^2$

$y = 0$ when $ax - bx^2 = 0$

$x = 0, a/b$

From figure $\frac{a}{b} = 1$

$$\boxed{a = b}$$

$\therefore y = ax - ax^2$

$$\frac{dN}{du} = a(u - u^2)$$

Total no. of particles,

$$N = \int dN$$

$$N = \int_0^1 a(u - u^2)\,du$$

$$N = \left[a\frac{u^2}{2} - a\frac{u^3}{3}\right]_0^1$$

$$N = \frac{a}{2} - \frac{a}{3} = \frac{a}{6} \Rightarrow \boxed{a = 6N}$$

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i) $V_{rms} = \sqrt{\dfrac{\int_0^1 u^2 \cdot dN}{\int_0^1 dN}} = \sqrt{\dfrac{1/2010}{1/603}} = \sqrt{\dfrac{3}{10}} \times 10^2 \text{ m/s}$

ii) $V_{avg} = \dfrac{\int_0^1 u \cdot dN}{\int_0^1 dN} = \dfrac{\alpha \int_0^1 u(u-u^2)\,du}{\alpha \int_0^1 u(u-u^2)\,du}$

$= \dfrac{1/12}{1/6} = 1/2 \times 10^2 \text{ m/s}$

iii) For $V = V_{mp}$

$$\frac{d}{du}(f(u)) = 0 \Rightarrow \frac{d}{du}(u-u^2) = 0$$

$$1 - 2u = 0$$

$$\boxed{u = 1/2}$$

$$\therefore V_{mp} = \frac{1}{2} \times 10^2 \text{ m/s}$$

$$V_{mp} = 50 \text{ m/s}$$

iv) $K.E = \frac{1}{2} m v_{rms}^2$

$= \frac{1}{2} \times 10 \times 10^{-3} \times \frac{3}{10} \times 10^4$

$= 15 \text{ J}$

Ex: Graph shows a hypothetical speed distribution for a sample of N gas particles $\left[\text{for } v > v_0, \frac{dN}{dv} = 0\right]$

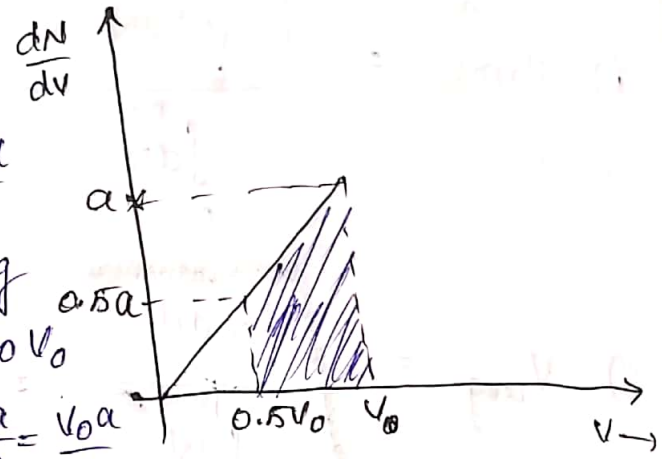
Calculate i) Total no. of particles

ii) no. of particles having speeds in b/w $0.5 v_0$ & v_0.

iii) V_{rms}

iv) V_{avg}

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① No. of particles

$$N_0 = \text{area} = \frac{1}{2} \times V_0 a = \frac{V_0 a}{2}$$

② No. of particles having speeds in blw $0.5V_0$ to V_0

$$N' = \text{area} = \frac{1}{2} \times \frac{V_0}{2} \times \frac{a}{2} = \frac{V_0 a}{8}$$

$$N = N_0 - N' = \left(\frac{1}{2} - \frac{1}{8}\right) V_0 a = \frac{3}{4} V_0 a.$$

③ eqn of st.line,

$$y = mx$$

$$\frac{dN}{dv} = \frac{aV}{V_0}$$

$$V_{rms} = \sqrt{\frac{\int_0^{V_0} v^2 \cdot dv}{\int_0^{V_0} dN}} = \frac{V_0}{\sqrt{2}}$$

④ $$V_{avg} = \frac{\int_0^{V_0} V \cdot dN}{\int_0^{V_0} dN} = \frac{2V_0}{3}$$

Ex: U are given particles of diff grps n_i represents no. of particles with speeds V_i. Find ① V_{mp} ② V_{avg} ③ V_{rms}.

Data:

n_i	V_i (m/s)
2	1
4	2
9	3
5	4
3	5

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① $V_{mp} = 3 m/s$

② $V_{avg} = \frac{2+8+27+20+15}{6+9+8}$

$= \frac{10+20+42}{14+9} = \frac{72}{23} = 3.13$

③ $V_{rms} = \sqrt{\frac{N_1V_1^2 + N_2V_2^2 + N_3V_3^2 + N_4V_4^2 + N_5V_5^2}{N_1+N_2+N_3+N_4+N_5}}$

$= \sqrt{\frac{2\times1 + 16 + 81 + 80 + 75}{23}}$

$= \sqrt{\frac{161+75+18}{23}} = \sqrt{\frac{161+93}{23}}$

$= \sqrt{\frac{254}{23}} = 3.32$

Degree of Freedom:

The term degrees of freedom of a system represents the possible independent motions that a system can have.

(or) The total no. of independent ways in which system can possess energy.

Types of motions:

① Translation motion energy:

(3) possible independent motions

energies – $\frac{1}{2}mv_x^2$, $\frac{1}{2}mv_y^2$, $\frac{1}{2}mv_z^2$

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② Rotational Motion:

③ possible independent motions:

energies → $\frac{1}{2} I_x \omega_x^2$; $\frac{1}{2} I_y \omega_y^2$, $\frac{1}{2} I_z \omega_z^2$

③ Vibrational motion:

It occurs at "very high temp".

In vibrational along a particular direction.

② possible energies → $\frac{1}{2} \mu v^2$; $\frac{1}{2} k r^2$

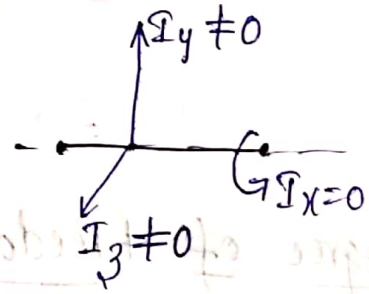
Mono-atomic gas: He, Ne, Ar, Kr, Xe, Rd etc

$f = 3$ (pure translational)

∵ Diatomic gas: H_2, O_2, N_2, etc

$f = 3 + 2 = 5$ (at room temp)

Trans rot



$f = 3 + 2 + 2 = 7$ (at very high temp)

Trans rot vib

Note:

$f_{vib} = 3N - 5$ (for linear

$= 3N - 6$ (for non-lin

Tri/polyatomic gas:

① linear Tri-atomic gas:

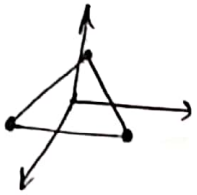


$f = 3 + 2 = 5$

Trans rot

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② Non-Linear tri-atomic gas:



$f = 3 + 3 = 6$

↓ Trans ↓ rot

Law of Equipartition of Energy:

According to this law, total energy is equally distributed among its various degrees of freedom.

we have,

Translational K.E of a molecule is $\frac{1}{2} m_0 v_{rms}^2$

$$K.E = \frac{1}{2} m_0 v_{rms}^2$$

$$\boxed{K.E = \frac{3}{2} KT}$$

∴ Energy associated with each molecule in each degree of freedom $= \frac{1}{2} KT$.

Energy associated with each molecule in f degrees of freedom $= \frac{f}{2} KT$

If gas contains N molecules, then energy associated $= \frac{f}{2} NKT$

If the gas contained is 1 mole then energy associated in each degree of freedom $= \frac{1}{2} N_A KT = \frac{1}{2} RT$

If energy associated with n-moles of gas having f-degrees of freedom $= \frac{f}{2} nRT$ — Interal Energy.

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$$U = \frac{f}{2} nRT$$

U depends on T.

For mixing up of gases:

① Total no. of moles is constant.

i.e, $n = n_1 + n_2 + \cdots$

② mass of gaseous mixture is constant.

$m = m_1 + m_2 + m_3 + \cdots$ here $n = \frac{m}{M}$

$$n M_{mix} = n_1 M_1 + n_2 M_2 + \cdots$$

$$M_{mix} = \frac{n_1 M_1 + n_2 M_2 + \cdots}{n_1 + n_2 + \cdots}$$

Equivalent degrees of freedom of mixture:-

N gases having degrees of freedom $f_1, f_2, \cdots f_N$ & moles $n_1, n_2, \cdots, n_N$ are mixed at same temp T.

Then internal energy of all gases before mixing;

$$U_i = U_1 + U_2 + \cdots + U_N$$

$$= \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT + \cdots + \frac{f_N}{2} n_N RT$$

and after mixing,

$$U_f = \frac{f_{eq}}{2} n_1 RT + \frac{f_{eq}}{2} n_2 RT + \cdots + \frac{f_{eq}}{2} n_N RT$$

Here, $U_i = U_f$

$$\frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT + \cdots + \frac{f_N}{2} n_N RT = \frac{f_{eq}}{2} n_1 RT + \cdots \frac{f_{eq}}{2} n_N$$

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$$f_{eq} = \frac{f_1 n_1 + f_2 n_2 + \cdots + f_N n_N}{n_1 + n_2 + \cdots + n_N}$$

Equivalent Temp of the mixture :-

If gases are mixed in a closed vessel at diff temp then there is no heat loss to outside & mixture finally attains same temp in eqm.

If N gases having degrees of freedom $f_1, f_2, \cdots f_N$ maintained at temp $T_1, T_2, \cdots T_N$ & moles $n_1, n_2, \cdots n_N$ are mixed in closed insulated vessel then

Internal enery of all gases before mixing,

$$U_i = U_1 + U_2 + \cdots U_N$$

$$= \frac{f_1}{2} n_1 R T_1 + \frac{f_2}{2} n_2 R T_2 + \cdots + \frac{f_N}{2} n_N R T_N$$

and after mixing,

$$U_f = \frac{f_1}{2} n_1 R T_f + \frac{f_2}{2} n_2 R T_f + \cdots + \frac{f_N}{2} n_N R T_f$$

Here, $U_i = U_f$

$$T_f = \frac{n_1 f_1 T_1 + n_2 f_2 T_2 + \cdots + n_N f_N T_N}{n_1 f_1 + n_2 f_2 + \cdots + n_N f_N}$$

Extra pts :

* Specific heat of gases :

① Specific heat of gas at const vol (c_v) :

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The amount of heat required to rise temp of unit mass of gas by 1°C at const volume.

mass of gas — gms

$$C_v = \frac{1}{m}\frac{dQ_v}{dT} \longrightarrow J\,kg^{-1}K^{-1}$$

If amount of gas is taken in "molar"

then, molar specific heat of gas,

$$C_v = \frac{1}{m}\frac{dQ_v}{dT} \longrightarrow J\,mol^{-1}K^{-1}$$

we have, $n = \frac{m}{M}$

$$C_v = \frac{M}{m}\frac{dQ_v}{dT}$$

$$\boxed{C_v = MC_v}$$

② Specific heat of gas at const pressure (C_p):

The amount of gas → gms

$$c_p = \frac{1}{m}\frac{dQ_p}{dT} \longrightarrow J\,kg^{-1}K^{-1}$$

amount of gas → mole

$$C_p = \frac{1}{n}\frac{dQ_p}{dT} \longrightarrow J\,mol^{-1}K^{-1}$$

$$\boxed{C_p = MC_p}$$

*Relation b/w (C_p & C_v) :-

$$c_p - c_v = R \qquad \gamma = \frac{c_p}{c_v}$$

$$MC_p - MC_v = R$$

$$\boxed{C_p - C_v = \gamma}$$

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again, $C_p - C_v = R$

$\gamma C_v - C_v = R$

$$\boxed{C_v = \frac{R}{\gamma - 1}} \quad \& \quad \boxed{C_p = \frac{\gamma R}{\gamma - 1}}$$

* we have,

$$U = \frac{f}{2} nRT$$

$$\Rightarrow dU = \frac{f}{2} nR\,dT \quad -(1)$$

and $dQ_v = nC_v\,dT \quad -(2)$

At const volume $V = \text{const}$

$dV = 0$

$pdV = 0$

$dW = 0$

I-law of Thermodynamics:

$dQ_v = dU + dW$ (dW → 0)

$dQ_v = dU$

$n C_v\,dT = \frac{f}{2} nR\,dT$

$$\boxed{C_v = \frac{fR}{2}}$$

& $C_p - C_v = R$

$C_p = R + C_v$

$C_p = R + \frac{fR}{2} \Rightarrow \boxed{C_p = \left(1 + \frac{f}{2}\right)R}$

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$$\gamma = \frac{C_p}{C_v}$$

$$\boxed{\gamma = 1 + \frac{2}{f}}$$

* for a mixture of gases

$$dU = dU_1 + dU_2 + \cdots$$

$$n C_{v_{mix}} dT = n_1 C_{v_1} dT + n_2 C_{v_2} dT + \cdots$$

$$\boxed{C_{v_{mix}} = \frac{n_1 C_{v_1} + n_2 C_{v_2} + \cdots}{n_1 + n_2 + \cdots}}$$

&

$$\boxed{C_{p_{mix}} = \frac{n_1 C_{p_1} + n_2 C_{p_2} + \cdots}{n_1 + n_2 + \cdots}}$$

$$\gamma_{mix} = \frac{C_{p_{mix}}}{C_{v_{mix}}}$$

(or)

$$n C_{v_{mix}} \cdot dT = n_1 C_{v_1} dT + \cdots$$

$$\frac{(n_1 + n_2 + \cdots) R}{\gamma_{mix} - 1} = \frac{n_1 R}{\gamma_1 - 1} + \frac{n_2 R}{\gamma_2 - 1} + \cdots$$

$$\boxed{\frac{n_1 + n_2 + \cdots}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1} + \cdots}$$

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Ex: Calculate

1) Avg K.E of Translation of O_2 at 27°C

2) Total K.E of O_2 at 27°C
molecule

3) Total K.E of 1 mole of oxygen at 27°C

Sol:

1) Avg K.E of Translation

$$= \frac{3}{2} K_b T = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300$$

$$= 9 \times 0.69 \times 10^{-21}$$

$$= 6.21 \times 10^{-21} \text{ J/molecule}$$

2) Total K.E of O_2 molecule

$$= \frac{5}{2} K_b T = \frac{5}{2} \times 1.38 \times 10^{-23} \times 300$$

$$= 15 \times 0.69 \times 10^{-21}$$

$$= 10.35 \times 10^{-21} \text{ J/molecule}$$

3) Total K.E of 1 mole O_2.

$$n = 1 \Rightarrow N = N_A$$

$$K.E = \frac{5}{2} \times N_A \times K_b T = 10.35 \times 10^{-21} \times 6.023 \times 10^{23}$$

$$= 62.338 \times 10^{2} \text{ J/mole}$$

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Ex: 15 gms of N_2 is enclosed in a vessel at a temp $T = 300K$. Find amount of heat required to double the rms velocity of these molecules.

$T_1 = 300K$, $M = 28$, $m = 15$ gms

$$n = \frac{m}{M} = \frac{15}{28}$$

we have,

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$$

$$T_2 = 4T_1$$

$$T_2 = 1200K$$

$$U_1 = \frac{f}{2} n RT = \frac{5}{2} \times \frac{15}{28} \times 8.314 \times 300$$

$$U_2 = \frac{5}{2} \times \frac{15}{28} \times 8.314 \times 1200$$

U = heat energy supplied $= U_2 - U_1$

$$= \frac{5}{2} \times \frac{15}{28} \times 8.314 \times 900$$

$$\boxed{U = 10021.35\ J}$$

Ex: 1 gms of He having rms velocity of molecules 1000 m/s & 4 gms of Oxygen having rms velocity 1000 m/s are mixed in a container which thermally insulated. what are rms velocity of He & O_2 molecules after eqm is attained.

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for He :-

$$V_1 = \sqrt{\frac{3RT_1}{M_1}} \Rightarrow \sqrt{\frac{3\times 8.314\times T_1}{4\times 10^{-3}}} = 1000$$

$$T_1 = 160.37\,K$$

$f_1 = 3$, $n_1 = \frac{m_1}{M_1} = \frac{1}{4}$

for O_2 :-

$$(1000)^2 = \frac{3\times 8.314\times T_2}{32\times 10^{-3}}$$

$$T_2 = 1282.98\,K$$

$f_2 = 5$, $n_2 = \frac{1}{8}$

eqm temp,

$$T_f = \frac{n_1 f_1 T_1 + n_2 f_2 T_2}{n_1 f_1 + n_2 f_2}$$

$$= \frac{\frac{1}{4}\times 3\times 160.37 + \frac{1}{8}\times 5\times 1282.98}{\frac{1}{4}\times 3 + \frac{1}{8}\times 5}$$

$$T_f = \frac{120.28 + 801.86}{0.75 + 0.625} = \frac{922.14}{1.375}$$

$$\boxed{T_f = 670.65\,K}$$

$$V_{He} = \sqrt{\frac{3\times 8.314\times 670.65}{4\times 10^{-3}}} \qquad V_{O_2} = \sqrt{\frac{3\times 8.314\times 670.65}{32\times 10^{-3}}}$$

$$= 2044.95\ m/s \qquad = 723.001\ m/s$$

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Ex: A cubical vessel of side 1m contains 1mole N_2 at a temp of 300K. If the molecules are assuming to move with rms velocity. Find

1) No. of collisions per sec. which the molecules may collide with the wall of the vessel.

2) Further if the vessel now thermally insulated with const velocity V is stopped suddenly and this results in the rise of temp of 2°C. Find V.

molecular density

$$n_0 = \frac{N}{V} = \frac{N_A}{1\times1\times1} = N_A = 6.023\times10^{23}/m^3$$

and $V_{rms} = \sqrt{\frac{3\times8.314\times300}{28\times10^{-3}}} = 516.948 m/s$

now, 1) $N = \frac{n_0 V_{rms}}{6}$

$$= \frac{6.023\times10^{23}\times516.948}{6}$$

$$\boxed{N = 518.93\times10^{23}}$$

extrapt

2) pressure of gas = $(\Delta P) \times \frac{\text{collision frequency}}{A}$

$$= m_0 V_{rms} \times \frac{n_0 V_{rms}}{6}$$

$$= \frac{m_0 n_0 V_{rms}^2}{3}$$

$$= \frac{28\times10^{-3}}{6.023\times10^{23}} \times \frac{6.023\times10^{23}}{3} \times (516.948)^2$$

$$P = 2494.195$$

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3) when container is suddenly stopped,

$$\frac{1}{2} m v^2 = \Delta U$$

└ mass of nitrogen gas $= 28 \times 10^{-3}$

$$\frac{1}{2} m v^2 = \frac{f}{2} n R \Delta T$$

$$\frac{1}{2} \times 28 \times 10^{-3} \times v^2 = \frac{5}{2} \times 1 \times 8.314 \times 2$$

$$\boxed{v = 54.49 \text{ m/s}}$$

Ex! An adiabatic vessel contains $n_1 = 3$ moles of diatomic gas. moment of inertia of each molecule is $I = 2.76 \times 10^{-46}$ kgm^2 & rms Ang velocity $(\omega_0) = 5 \times 10^{12}$ rad/sec another adiabatic vessel contains $n_2 = 5$ moles of mono atomic gas at a temp $y = 70$ K. Assume gases to be ideal. Calculate The rms Angular velocities of diatomic molecules when the 2 vessels are connected by a thin tube of negligible volume.

for diatomic molecule,

$$\frac{1}{2} I \omega_0^2 = 2 \times \frac{y}{2} K_b T_1$$

$$T_1 = \frac{2.76 \times 10^{-46} \times 25 \times 10^{24}}{2 \times 1.38 \times 10^{-23}}$$

$$= 25 \times 10$$

$$\boxed{T_1 = 250}$$

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$n_1 = 3$, $f_1 = 5$

for monoatomic gas,

$n_2 = 5$, $f_2 = 3$, $T_2 = 470K$

when mixed,

$$T_f = \frac{n_1 f_1 T_1 + n_2 f_2 T_2}{n_1 f_1 + n_2 f_2}$$

$$= \frac{3 \times 5 \times 250 + 5 \times 3 \times 470}{3 \times 5 + 5 \times 3}$$

$$\therefore \boxed{T_f = 360}$$

finally,

$$\frac{1}{2} I \omega^2 = 2 \times \frac{1}{2} k T_f$$

$$\frac{1}{2} \times 2.76 \times 10^{-46} \times \omega^2 = 1 \times 1.38 \times 10^{-23} \times 360$$

$$\omega^2 = 36 \times 10^{24}$$

$$\boxed{\omega = 6 \times 10^{12}}$$

Ex: A non-linear tri-atomic gas is filled inside a vessel. If α fraction of moles dissociate into individual atoms then find ~~of freed~~ avg degrees of freedom for the mixture. Neglect vibrational degrees of freedom.

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Let n is the total no. of moles of gas.

Therefore, no. of moles of mono-atomic gas formed: $n_1 = (n\alpha) \times 3$. $f_1 = 3$

no. of moles of diatomic gas formed.

$$n_2 = n - n\alpha \qquad f_2 = 6$$

$$f_{avg} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

$$= \frac{3\alpha + 6}{2\alpha + 1}$$

Ex: for a gas sample N_0 no. of molecules $N(v)$ is given as $N(v) = \frac{dN}{dv} = \left(\frac{3N_0}{V_0^3}\right) v^2$ for $0 \le v \le v_0$

$= 0$ for $v > v_0$

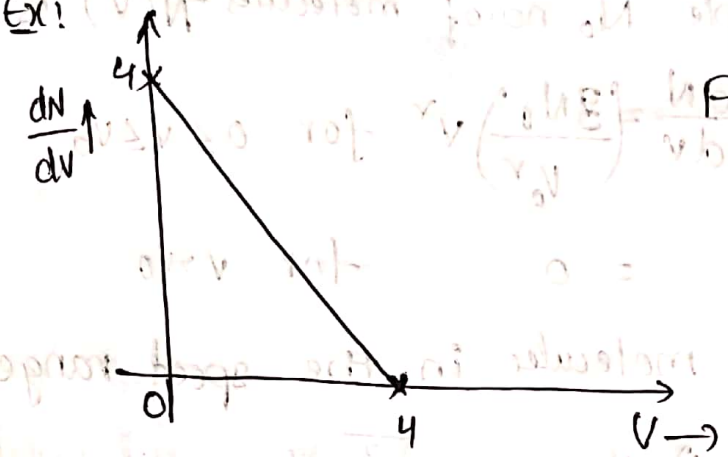
where dN = no. of molecules in the speed-range v to $v+dv$. Find 1) $v_{rms} = \sqrt{\frac{3}{5}} v_0$

2) v_{avg}

$$v_{rms} = \sqrt{\frac{\int v^2 . dN}{\int dN}}$$

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Ex:



Find 1) Vrms

2) Vavg

3) Total no of molecules.

Eqn of line,

$y = mx + c$

$$\frac{dN}{dv} = -1(v) + 4$$

$$\frac{dN}{dv} = 4 - v$$

$$N = 4v - \frac{v^2}{2}$$

$$V_{rms} = \sqrt{\frac{\int_0^4 v^2 . dN}{\int_0^4 dN}} = \sqrt{\frac{\int_0^4 v^2 (4-v) . dv}{\int_0^4 (4-v) . dv}}$$

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$$= \sqrt{\frac{\left[\frac{4v^3}{3} - \frac{v^4}{4}\right]_0^4}{\left[4v - \frac{v^2}{2}\right]_0^4}}$$

$$= \sqrt{\frac{\frac{4\times 4^3}{3} - \frac{4^4}{4}}{4^2 - \frac{4^2}{2}}} = \sqrt{\frac{4^4\left(\frac{1}{6}\right)}{4^2\left(\frac{1}{2}\right)}}$$

$$= \frac{4}{\sqrt{6}} = 2\sqrt{\frac{2}{3}} = \sqrt{\frac{8}{3}}$$

$$v_{avg} = \frac{\int_0^4 v \cdot dN}{\int_0^4 dN} = \frac{\int_0^4 v(4-v)\cdot dv}{\int_0^4 (4-v)\cdot dv}$$

$$= \frac{\left[4\frac{v^2}{2} - \frac{v^3}{3}\right]_0^4}{\left[4v - \frac{v^2}{2}\right]_0^4}$$

$$= \frac{4^3\left(\frac{1}{6}\right)}{4^2\left(\frac{1}{2}\right)}$$

$$\boxed{v_{avg} = 4/3}$$

$$N = 4\times 4 - \frac{4\times 4}{2} = 16\times\frac{1}{2} = 8$$

* Two bodies in thermal eqm will have const average speed but rms speed may be different.

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Ex! A vessel contains a mixture 7 gms of N_2, 11 gms of CO_2 at 290K. If pressure of mixture is 1 atm. Calculate its density.

$$PV = nRT$$

$$10^5 V = \left(\frac{7}{28} + \frac{11}{44}\right) \times 8.314 \times 290$$

$$V = \frac{8.314 \times 290}{2} \times 10^{-5}$$

$$= 8.314 \times 145 \times 10^{-5}$$

$$= 1205.53 \times 10^{-5}$$

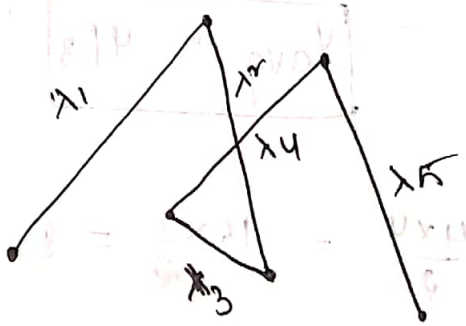
$$d = \frac{Tm}{TV} = \frac{18 \times 10^{-3}}{8.314 \times 145 \times 10^{-5}} = 1.49 \text{ kg/m}^3$$

$$\boxed{d = 1.5 \text{ kg/m}^3}$$

Mean free path :

All molecules of gas are moving in random called as brownian movement.





① mean free path :

$$\lambda = \frac{\lambda_1 + \lambda_2 + \cdots}{N}$$

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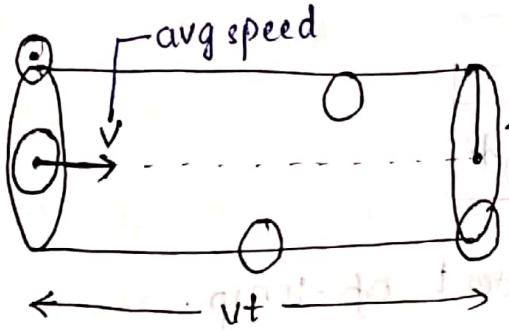
② The time interval b/w 2 successive collisions is called "relaxation time"

③ No. of collisions per second of molecules is called collision frequency. $\boxed{\Delta f = \frac{1}{t}}$

Derivation:

Assumption –

Only one molecule is in motion and all other are at rest.



$\sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8KT}{\pi m_0}}$

$n_0 = \frac{N}{V}$ = molecular density.

no. of molecules in a cylinder

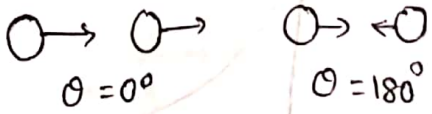
$= \pi d^2 v t \times n_0$

$$\text{collision frequency } (f) = \frac{\pi d^2 v t}{t} \times n_0$$

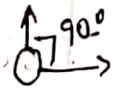
But actually molecules moves at random.

$$\Delta f = \pi d^2 v_{rel} n_0$$

$$\boxed{\Delta f = \pi n_0 d^2 \sqrt{2} v}$$



avg: 90°



$v_{rel} = \sqrt{v^2 + v^2} = \sqrt{2} v$

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Relaxation time,

$$\tau = \frac{1}{\Delta f}$$

$$\boxed{\tau = \frac{1}{\sqrt{2}\pi n_0 d^2 v}}$$

$$v = \sqrt{\frac{8RT}{\pi M}}$$

as temp (↑), v(↑)

$v \propto \sqrt{T}$

$\Rightarrow \tau(\downarrow)$.

mean free path,

$$\lambda = v\tau$$

$$\boxed{\lambda = \frac{1}{\sqrt{2}\pi n_0 d^2}}$$

i.e, λ is independent of temp.

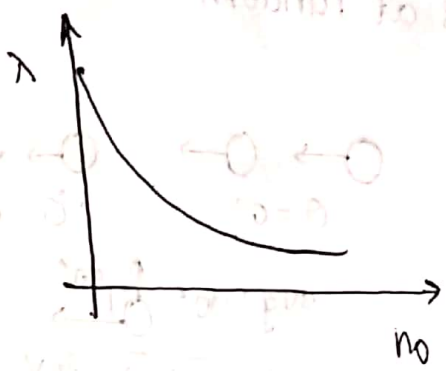
again, $n_0 = \frac{N}{V} = \frac{P}{RT}$

$$\boxed{\lambda = \frac{RT}{\sqrt{2}\pi d^2 P}}$$

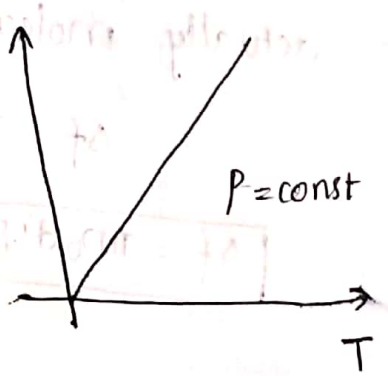
and $\boxed{\lambda = \frac{m_0}{\sqrt{2}\pi \rho d^2}}$

Graphs:

1)

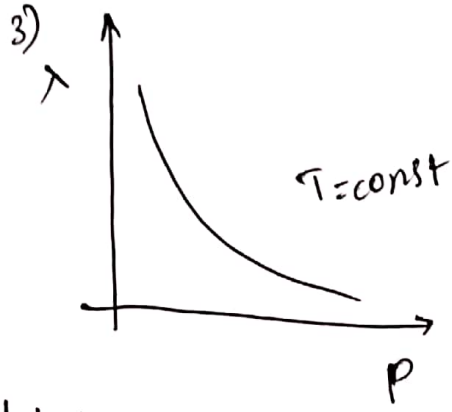


2)

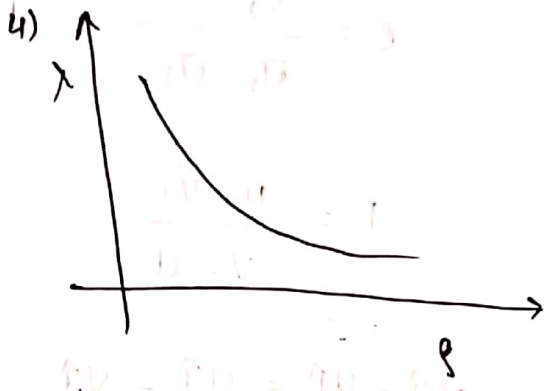


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3)



4)



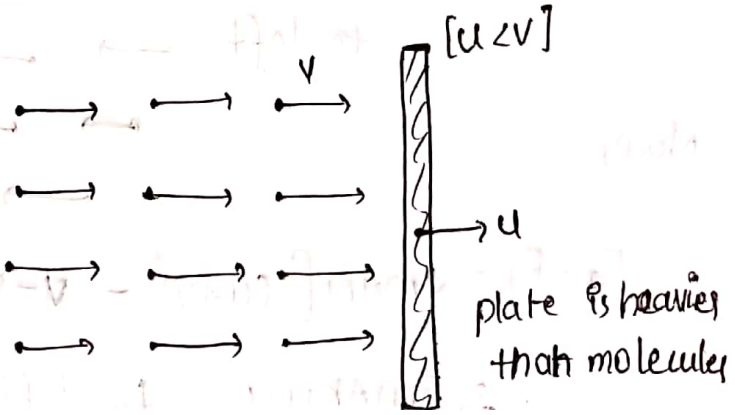
Notes

$f_{vib} = 3N-5$ (for linear atoms)

$f_{vib} = 3N-6$ (for non-linear atoms)

Ex: A beam of gas molecules is incident normally on plate AB. Each gas molecules has mass m & velocity 'v'. The incident beam falls on the plate of area 'A' & molecules collide with plate elastically. No. of molecules in unit volume of beam is 'n' and the plate is moved to right a force 'F_1' is needed to keep it moving with const velocity 'u' ($<v$) when the plate is moved to left with const velocity u. an external force 'F_2' is needed. Find $F_2 - F_1$.

Sol: case(1):



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$$e = \frac{\bar{V}_2 - \bar{V}_1}{\bar{u}_2 - \bar{u}_1}$$

$$1 = \frac{u - \bar{V}_1}{V - u}$$

$$V\hat{\uparrow} - u\hat{\uparrow} = u\hat{\uparrow} - V_1\hat{\uparrow}$$

$$\bar{V}_1\hat{\uparrow} = 2u\hat{\uparrow} - V\hat{\uparrow}$$

Change in momentum of molecule

= final - initial

$= m(2u - V)\hat{\uparrow} - mV\hat{\uparrow}$

$= 2m(V - u)(\hat{-\uparrow})$

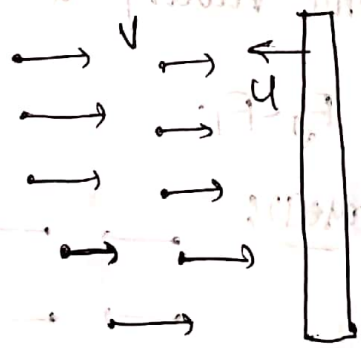
Force on plate = Change in momentum × collision frequency

$= 2m(V-u) \cdot A(V-u)n$

$= 2m(V-u)^2 An\,(-\hat{\uparrow})$

$\therefore F_1 = 2m(V-u)^2 An$ to left.

Case (2):



$F_2 = 2mAn(u+V)^2$ to left

Now,

$F_2 - F_1 = 2mAn\left[(u+V)^2 - (V-u)^2\right]$

$= 8mAnuV$ to left.

Scanned with CamScanner